

Unit 15

HALOGEN COMPOUNDS

74 items · 170 marks

Syllabus sections in this unit

15.1 Halogenoalkanes

Unit 15: Halogen compounds

1. M/J 26/P21/Q5(e)

Context

- 5** Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

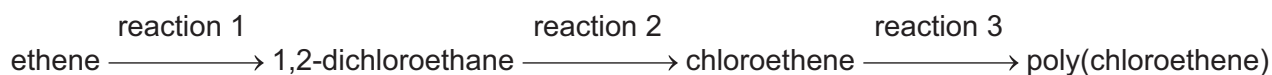


Figure 5.1

- (e) When 1,2-dichloroethane is heated in reaction 2, thermal decomposition occurs. The reaction is endothermic.

State the type of reaction that occurs.

[1]

2. MAR 26/P22/Q3(a)

Context

- 3** Secondary halogenoalkanes are important building blocks in organic chemistry. 2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane

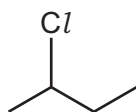


Figure 3.1

- (a) Explain why 2-chlorobutane is classified as a secondary halogenoalkane.

[1]

3. MAR 26/P22/Q3(d)

Context

3 Secondary halogenoalkanes are important building blocks in organic chemistry.

2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane

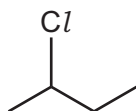


Figure 3.1

(d) The rates of hydrolysis of three secondary halogenoalkanes, **B**, **C** and **D**, are compared.

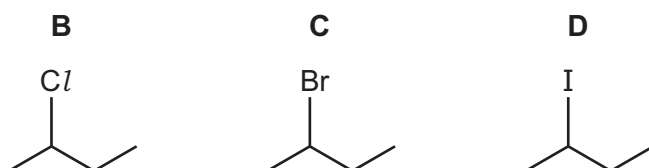


Figure 3.3

Each halogenoalkane is dissolved in ethanol and reacted with $\text{AgNO}_3(\text{aq})$.

Identify the order of the rates of hydrolysis of compounds **B**, **C** and **D**. Explain your answer.

fastest > > slowest

explanation

[2]

4. MAR 26/P22/Q4(c)

Context

4 Cyanoethanoic acid, $\text{CH}_2(\text{CN})\text{COOH}$, is a weak organic acid.

(c) Figure 4.3 shows a synthesis of $\text{CH}_2(\text{CN})\text{COOH}$.

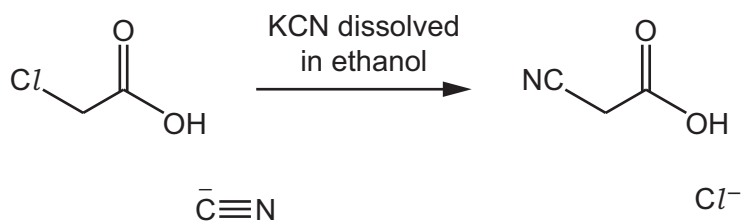


Figure 4.3

(i) The reaction mechanism of the synthesis in Figure 4.3 has a single step.

Complete Figure 4.3 to show this mechanism. Include lone pairs, charges, dipoles and curly arrows as appropriate.

[2]

(ii) Identify the nucleophile in Figure 4.3.

[1]

(iii) Identify the mechanism of the reaction in Figure 4.3.

[1]

5. O/N 25/P22/Q4(b)

Context

4 Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$.

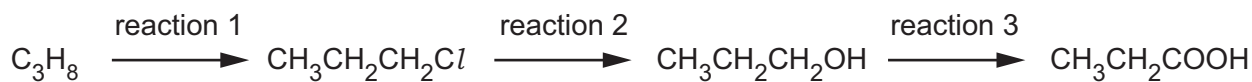


Fig. 4.1

(b) State the reagent and solvent required for reaction 2.

[1]

6. O/N 25/P24/Q3(b)(ii)-(iii)

Context

- 3 Propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

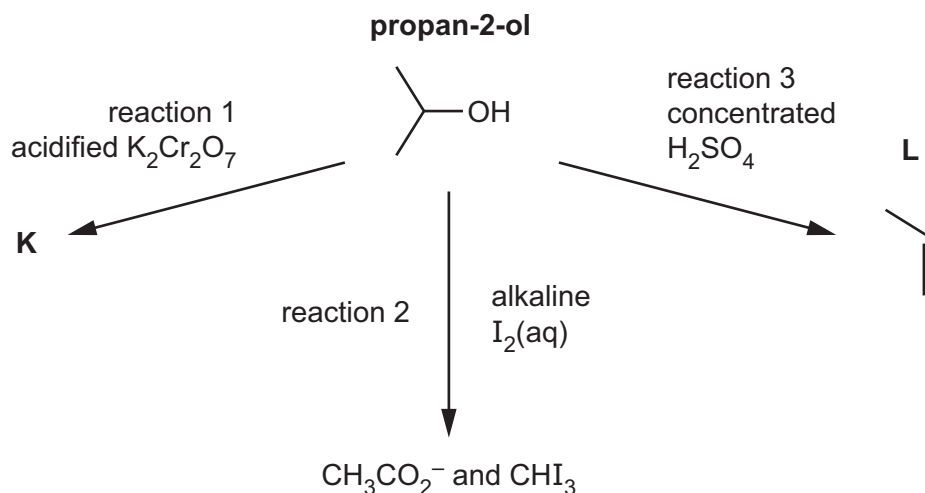


Fig. 3.1

Context: part (b) (introduction)

- (b) Propan-2-ol reacts with sodium to produce $(\text{CH}_3)_2\text{CH-O}^-$ anions.

These anions react with 2-bromopropane to form compound **N**, as shown in Fig. 3.3.

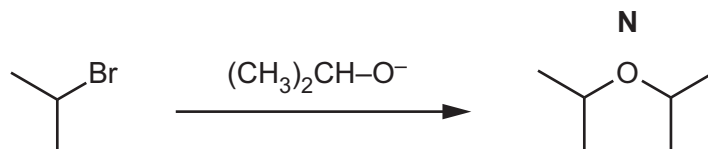


Fig. 3.3

- (ii) The reaction of 2-bromopropane with $(\text{CH}_3)_2\text{CH-O}^-$ anions follows an $\text{S}_{\text{N}}1$ mechanism.

Complete Fig. 3.4 to show this mechanism. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 3.4

- (iii) Suggest how the rate of the $\text{S}_{\text{N}}1$ reaction would change, if at all, if 2-chloropropane were used instead of 2-bromopropane.

Explain your answer.

[3]

[2]

7. O/N 25/P24/Q4(b)(i)

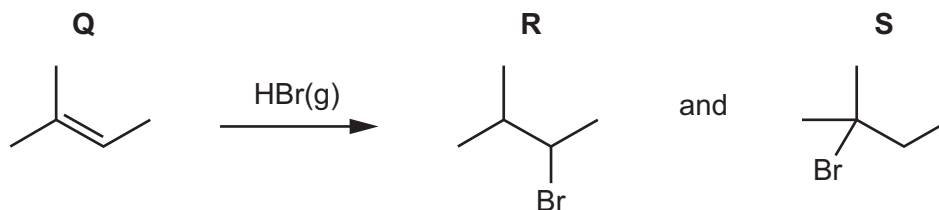
Context

- 4** Compounds **P** and **Q** are structural isomers.

**Fig. 4.1**

Context: part (a)(v)

- (v)** **Q** reacts with HBr(g) to produce two structural isomers, **R** and **S**, as shown in Fig. 4.2.

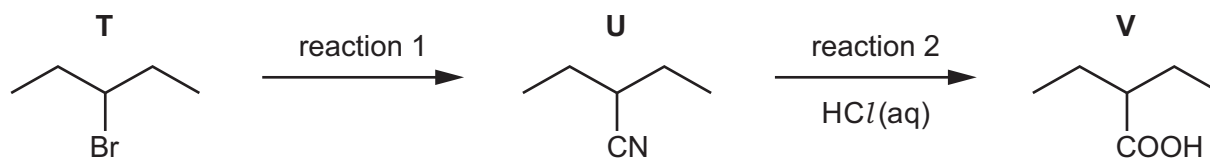
**Fig. 4.2**

State and explain why isomer **S** is the major product of the reaction.

[2]

Context: part (b) (introduction)

- (b)** Fig. 4.3 shows a synthesis starting from **T**, a different isomer of **R** and **S**.

**Fig. 4.3**

- (i)** Identify the reagent and conditions for reaction 1.

[1]

8. M/J 25/P22/Q4(b)(ii)

Context: part (b) (introduction)

- (b)** Propan-1-ol is converted to compound **T** in a three-step synthesis.

In step 1, SOCl_2 is added to propan-1-ol to produce compound **E**.

In step 2, **E** reacts with a suitable reagent to produce butanenitrile.

In step 3, butanenitrile is heated with NaOH(aq) .

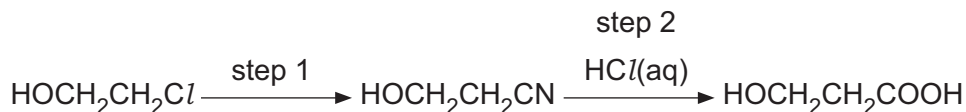
- (ii)** Identify the reagent and conditions required in step 2.

[1]

9. M/J 25/P23/Q6(c)(i)

Context: part (c) (introduction)

- (c) $\text{HOCH}_2\text{CH}_2\text{Cl}$ reacts in a two-step synthesis to produce $\text{HOCH}_2\text{CH}_2\text{COOH}$, as shown in Fig. 6.2.

**Fig. 6.2**

- (i) State the reagent and conditions for step 1.

[1]

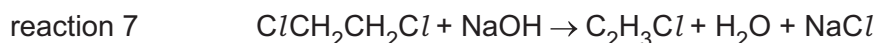
10. MAR 25/P22/Q1(c)(ii)

Context

- 1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

Context: part (c) (introduction)

- (c) 1,2-dichloroethane, $\text{ClCH}_2\text{CH}_2\text{Cl}$, reacts with NaOH to produce an unsaturated compound, $\text{C}_2\text{H}_3\text{Cl}$, as shown in reaction 7.



- (ii) State the conditions for reaction 7.

[1]

11. O/N 24/P21/Q2(c)(iv)

Context

- 2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

- (c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO .

NaClO is made from chlorine in a disproportionation reaction.

- (iv) Aqueous AgNO_3 dissolved in ethanol reacts with an aqueous solution of CHCl_3 .

State what is observed in this reaction. Explain your answer.

[2]

12. O/N 24/P23/Q2(c)(iv)

Context

- 2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

- (c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO .

NaClO is made from chlorine in a disproportionation reaction.

- (iv) Aqueous AgNO_3 dissolved in ethanol reacts with an aqueous solution of CHCl_3 .

State what is observed in this reaction. Explain your answer.

[2]

13. M/J 24/P22/Q4(b)

Context

4 $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ exists as a pair of stereoisomers.

(b) A sample of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ reacts with NaOH to make $\text{CH}_3(\text{CH}_2)_5\text{CH(OH)CH}_3$ in an $\text{S}_{\text{N}}1$ mechanism.

Complete Fig. 4.1 to show the mechanism for the reaction of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ and NaOH .

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 4.1**

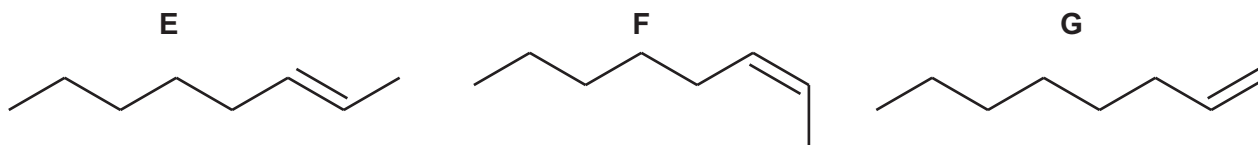
[3]

14. M/J 24/P22/Q4(d)

Context

4 $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ exists as a pair of stereoisomers.

(d) $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ is heated with **D** to produce three different molecules, **E**, **F** and **G**.

**Fig. 4.2**

(i) Name the type of reaction.

[1]

(ii) Identify **D** and the conditions used.

[1]

15. M/J 24/P23/Q4(c)

(c) Substitution reactions of NH_3 and OH^- with halogenoalkanes both involve a lone pair of electrons.

(i) Name the role of NH_3 and OH^- in these reactions.

[1]

(ii) Suggest which species, NH_3 or OH^- , is more reactive during these reactions. Explain your answer.

[1]

16. M/J 24/P23/Q4(d)

- (d) When 2-bromo-2-methylpropane reacts with OH^- , two mechanisms, $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$, both occur. The $\text{S}_{\text{N}}2$ mechanism has a slower rate.

Fig. 4.1 shows the reaction pathway diagram for the $\text{S}_{\text{N}}1$ mechanism.

Sketch a graph on Fig. 4.1 to show the reaction pathway for the $\text{S}_{\text{N}}2$ mechanism.

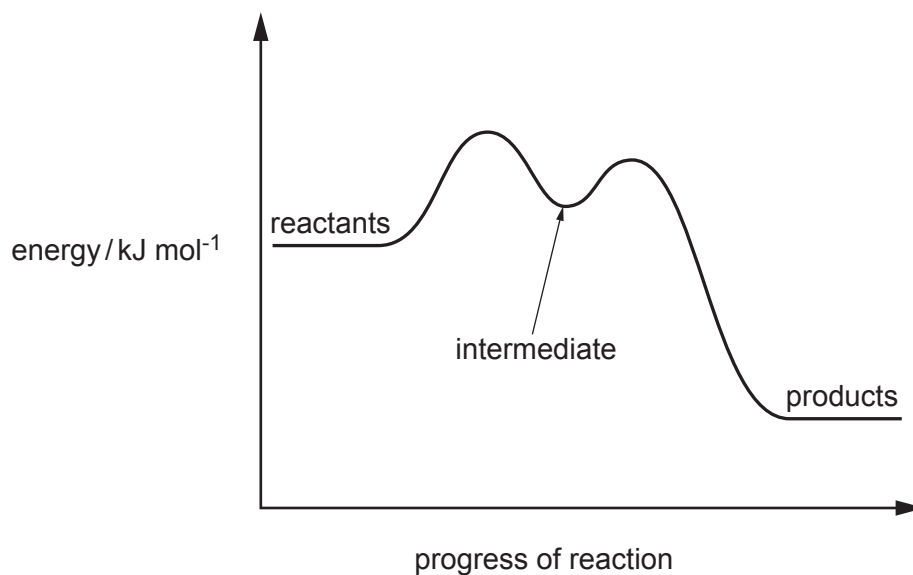
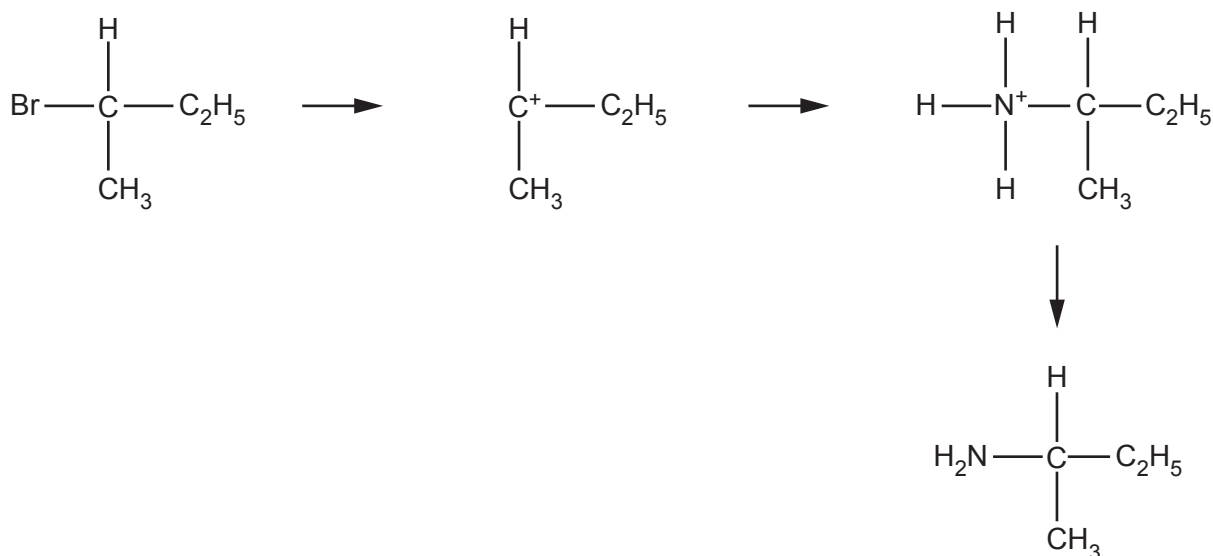


Fig. 4.1

[2]

17. M/J 24/P23/Q4(e)(i)-(ii)

- (e) (i) Complete Fig. 4.2 to show the mechanism for the S_N1 reaction that occurs when $\text{CH}_3\text{CHBrC}_2\text{H}_5$ reacts with NH_3 to produce $\text{CH}_3\text{CH}(\text{NH}_2)\text{C}_2\text{H}_5$. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 4.2**

[3]

- (ii) Identify the inorganic product that forms in the reaction in Fig. 4.2.

[1]

18. M/J 24/P23/Q4(f)

- (f) (i) Complete Table 4.1 by drawing the structural formula of the intermediate that is formed when 2-bromo-2-methylpropane reacts in an S_N1 reaction.

Table 4.1

	2-bromobutane	2-bromo-2-methylpropane
structural formula of intermediate in S_N1 reaction	$\begin{array}{c} \text{H} \\ \\ \text{C}^+ - \text{C}_2\text{H}_5 \\ \\ \text{CH}_3 \end{array}$	

[1]

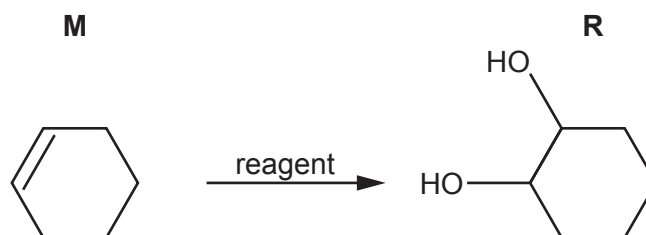
- (ii) Identify the halogenoalkane in Table 4.1 that has the greater tendency to react using the S_N1 mechanism. Explain your answer.

[2]

19. M/J 24/P23/Q5(b)(i)

Context: part (a)

- 5 (a)** **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

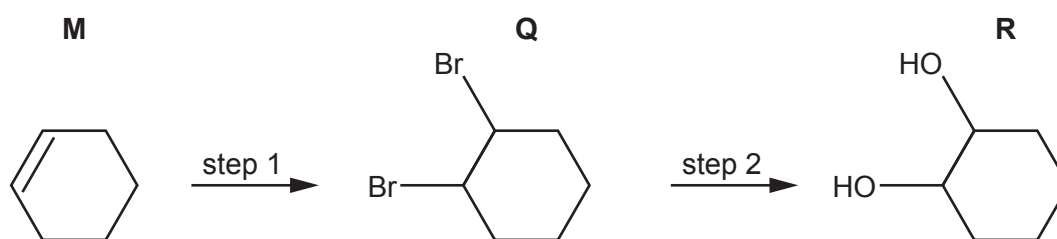
**Fig. 5.1**

Identify the reagent and conditions for this reaction.

[1]

Context: part (b) (introduction)

- (b)** **R** is also made from **M** by two steps, as shown in Fig. 5.2.

**Fig. 5.2**

- (i)** Identify the reagents and conditions for steps 1 and 2 in Fig. 5.2.

step 1

step 2

[2]

20. MAR 24/P22/Q2(d)(ii)

Context

2 Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

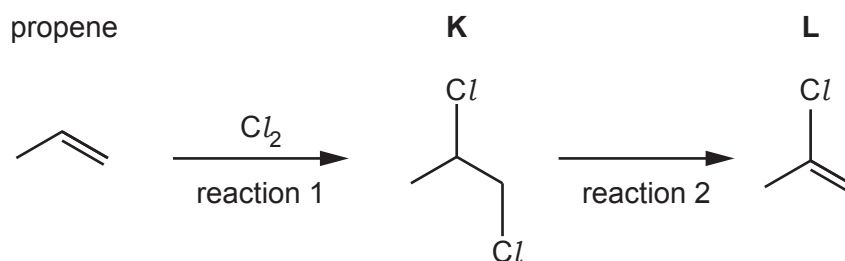
Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

Context: part (d) (introduction)

(d) Cl_2 reacts readily with propene to form **K**, 1,2-dichloropropane.

K can be used to form **L**.

**Fig. 2.1**

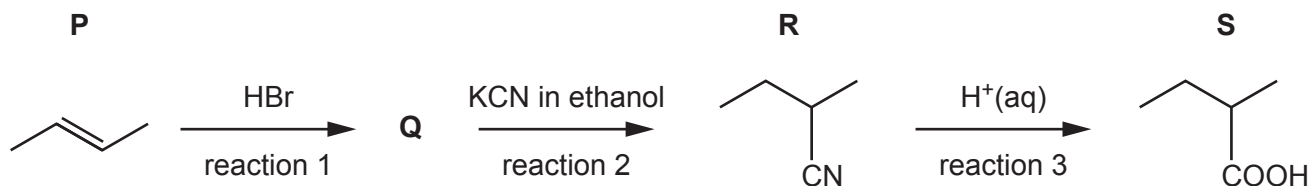
(ii) Identify the reagent and conditions for reaction 2.

[1]

21. MAR 24/P22/Q4(b)(ii)

Context

- 4** Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

**Fig. 4.1**

- (ii) Name the mechanism in reaction 2.

[1]

22. M/J 23/P21/Q3(e)

Context

- 3** Chlorine is a very reactive element.

(e) CHCl_3 and HF are used to form CHClF_2 in a substitution reaction.

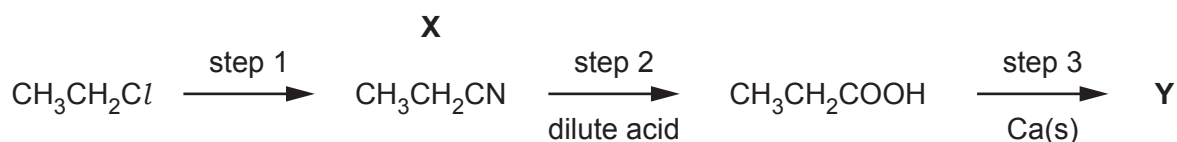
Construct an equation for this reaction.

[1]

23. M/J 23/P22/Q3(a)(i)

Context

- 3** Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

- (a) (i) State the reagent and conditions for step 1 in Fig. 3.1.

[1]

24. MAR 23/P22/Q4(a)(i)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

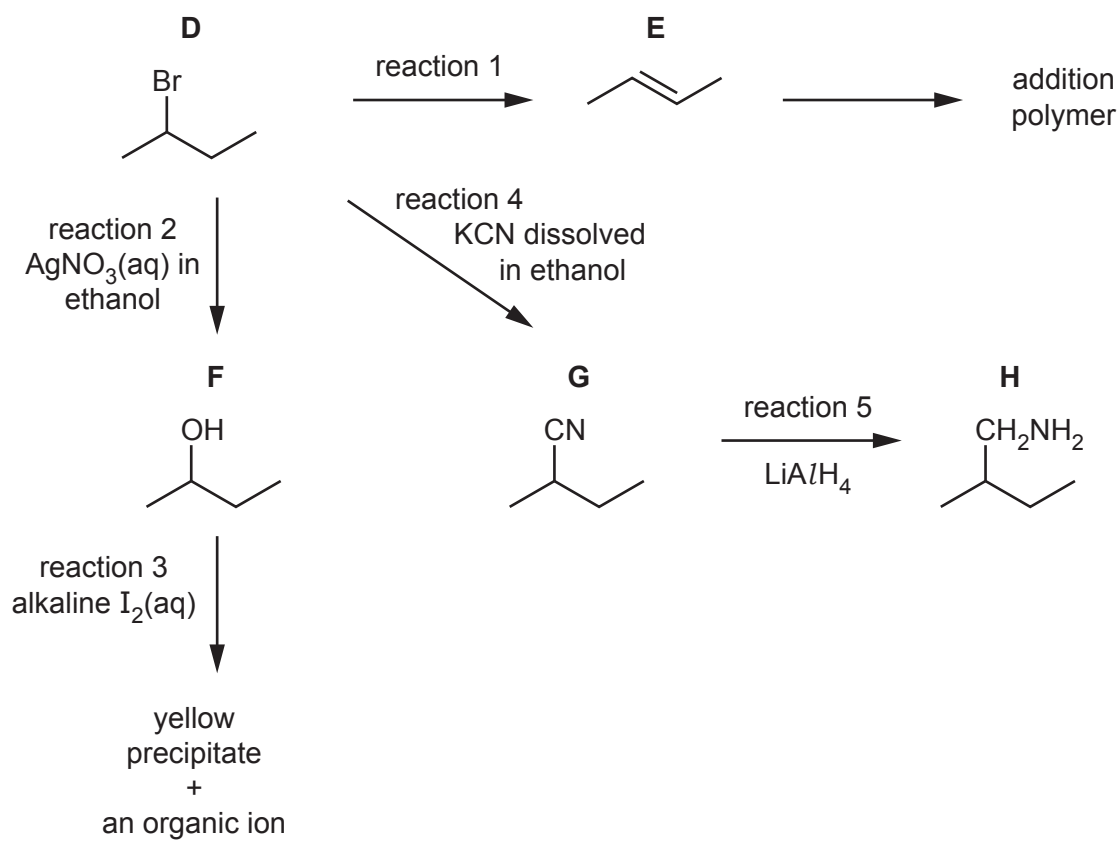


Fig. 4.1

(a) (i) State the reagent and conditions used to form **E** in reaction 1.

[1]

25. MAR 23/P22/Q4(b)(i)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

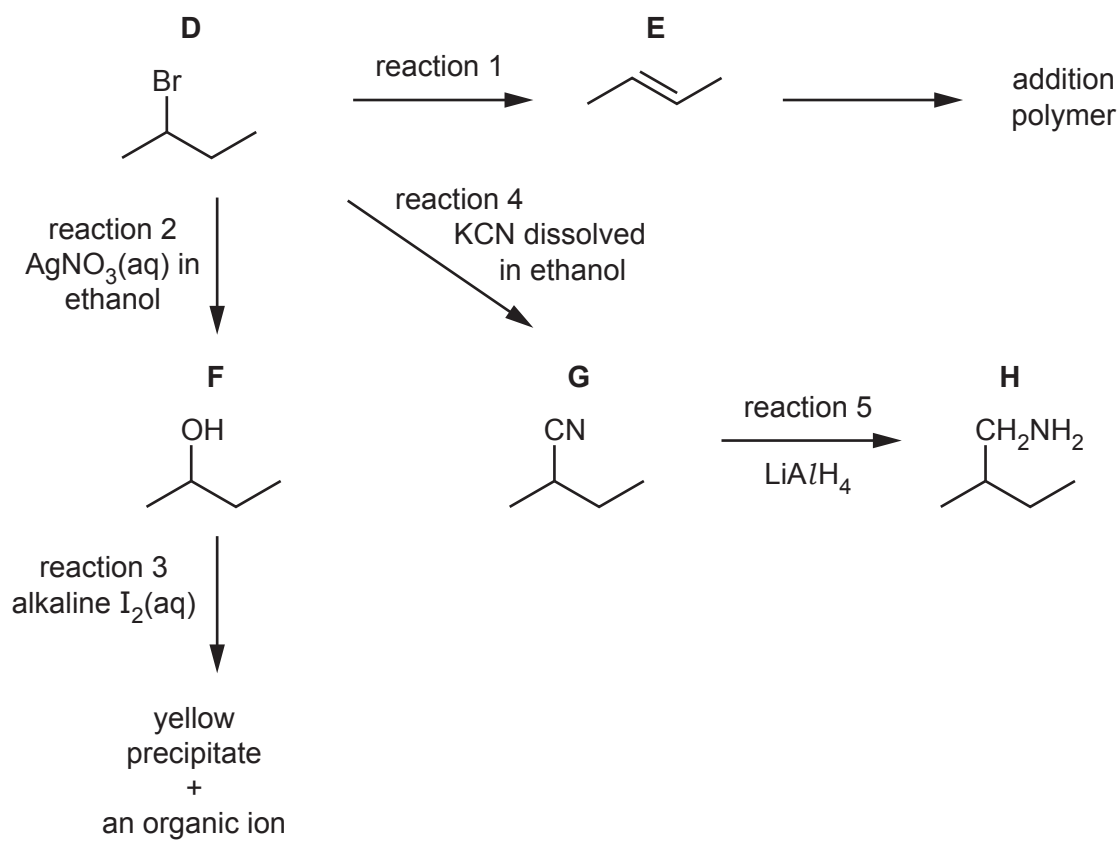


Fig. 4.1

(b) (i) Predict what is observed in reaction 2.

[1]

26. MAR 23/P22/Q4(c)(i)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

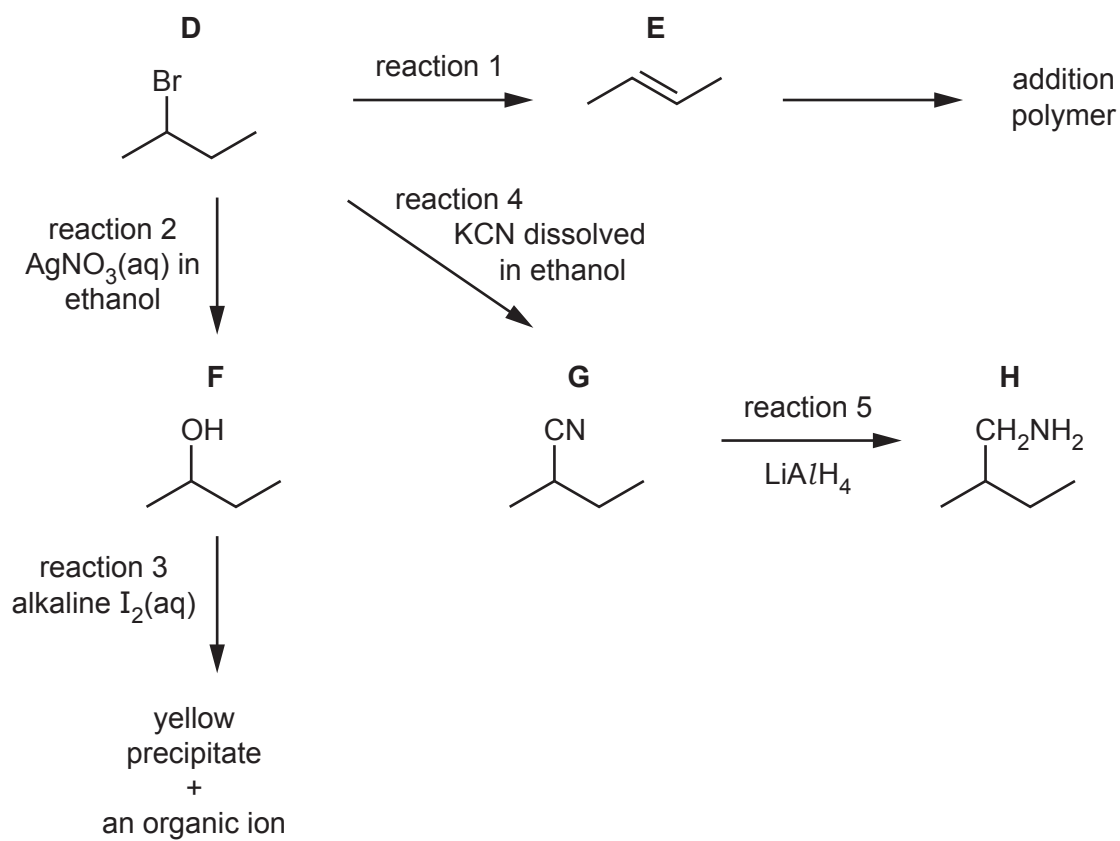


Fig. 4.1

(c) (i) State the type of reaction that occurs in reaction 4.

[1]

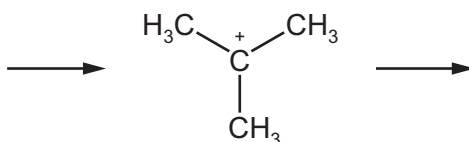
27. O/N 22/P22/Q1(e)

Context

1 Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

(e) $\text{OH}^-(\text{aq})$ reacts with 2-bromo-2-methylpropane in an $\text{S}_{\text{N}}1$ reaction. The molecular ion $(\text{CH}_3)_3\text{C}^+$ forms as the intermediate in this reaction.

(i) Draw the mechanism for the S_N1 reaction of OH^- with 2-bromo-2-methylpropane. Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate. Draw the structures of the organic reactant and organic product.



[3]

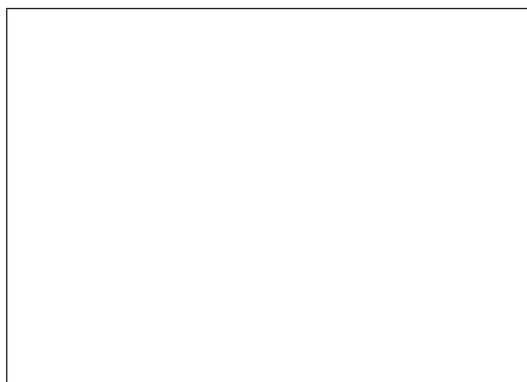
(ii) 2-bromo-2-methylpropane is a tertiary bromoalkane.

Define tertiary bromoalkane.

[1]

(iii) Organic compound **M** forms when 2-bromo-2-methylpropane is heated with **ethanolic** OH^- .

Draw the structure of **M**.

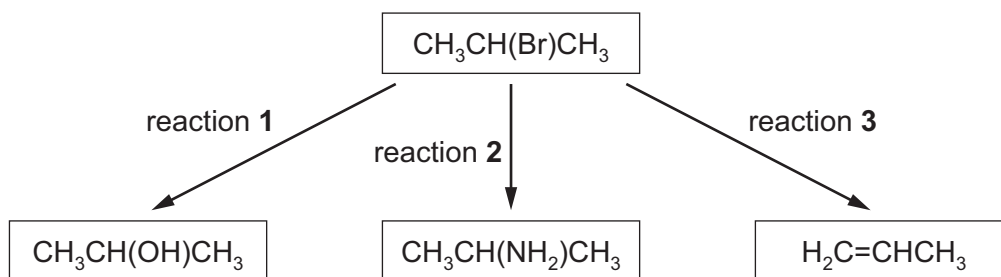


[1]

28. M/J 22/P22/Q5(a)

Context

5 Fig. 5.1 shows three reactions of 2-bromopropane, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$.

**Fig. 5.1**

(a) Complete Table 5.1 for each reaction, by:

- stating the reagent and conditions used
- identifying the type of reaction that occurs.

Table 5.1

reaction	reagent and conditions	type of reaction
1		
2		
3		

[6]

29. M/J 22/P22/Q5(b)

Context

5 Fig. 5.1 shows three reactions of 2-bromopropane, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$.

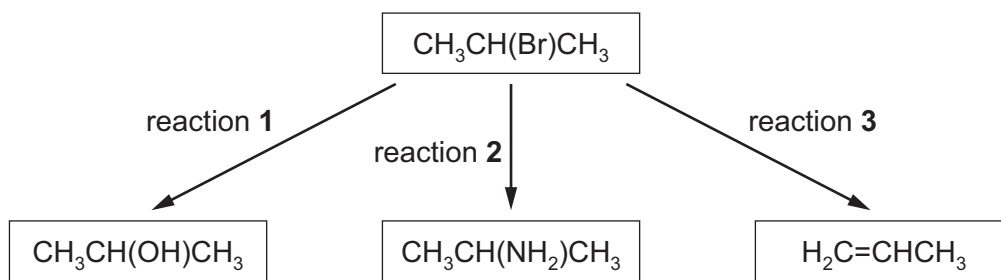


Fig. 5.1

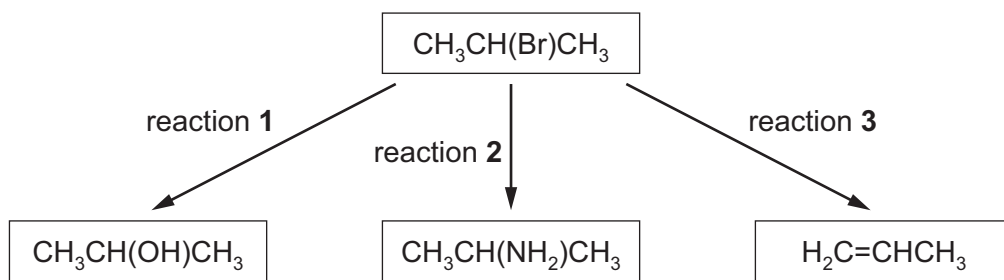
(b) A sample of 2-iodopropane, $\text{CH}_3\text{CH}(\text{I})\text{CH}_3$, reacts under the same conditions as reaction **1** to produce $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$.

Explain why 2-iodopropane reacts at a faster rate than 2-bromopropane.

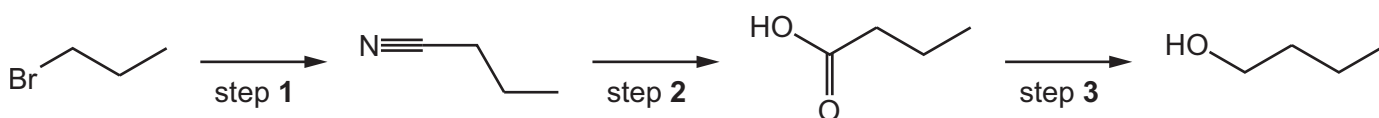
[2]

30. M/J 22/P22/Q5(c)(i)

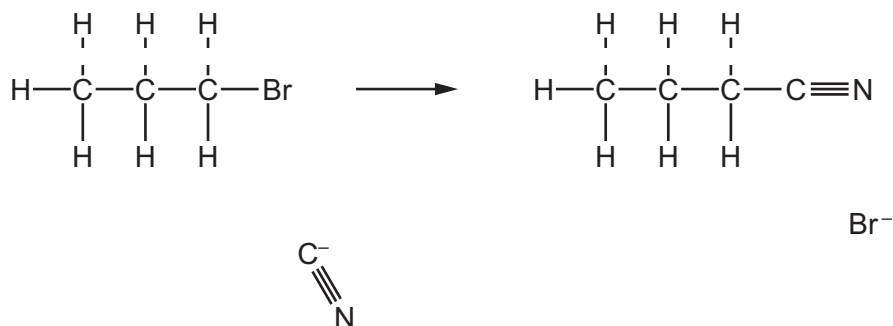
Context

5 Fig. 5.1 shows three reactions of 2-bromopropane, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$.**Fig. 5.1**

Context: part (c) (introduction)

(c) Fig. 5.2 shows how butan-1-ol can be made from 1-bromopropane in three steps.**Fig. 5.2****(i)** In step 1, 1-bromopropane reacts with CN^- to form butanenitrile.

Complete Fig. 5.3 to show the mechanism for step 1. Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.

**Fig. 5.3**

[2]

31. M/J 22/P23/Q4(b)(i)-(iii)

Context: part (b) (introduction)

- (b) Two bottles labelled **Q** and **M** each contain a straight-chain halogenoalkane with molecular formula C_4H_9X , where **X** represents Cl, Br or I.
A sample from each bottle is added to separate samples of equal amounts of aqueous silver nitrate in ethanol. In each reaction, the same organic product, **T**, and a precipitate are made, as shown in Fig. 4.2.

**Fig. 4.2**

Table 4.1 describes the colour of each of the precipitates made.

Table 4.1

halogenoalkane added to $\text{AgNO}_3(\text{aq})$ in ethanol	colour of precipitate
Q	white
M	yellow

- (i) Identify the functional group present in **T** and name the type of reaction that occurs using the information in Fig. 4.2 and Table 4.1.

functional group in **T**

type of reaction

[2]

- (ii) Construct an ionic equation to describe the formation of the yellow precipitate produced when **M** reacts with $\text{AgNO}_3(\text{aq})$ in ethanol.

[1]

- (iii) Describe which reagent, **Q** or **M**, will produce a precipitate more quickly when each is added to $\text{AgNO}_3(\text{aq})$ in ethanol. Explain your answer.

reagent

[1]

32. M/J 22/P23/Q4(b)(v)

Context: part (b) (introduction)

- (b) Two bottles labelled **Q** and **M** each contain a straight-chain halogenoalkane with molecular formula C_4H_9X , where **X** represents Cl, Br or I.
A sample from each bottle is added to separate samples of equal amounts of aqueous silver nitrate in ethanol. In each reaction, the same organic product, **T**, and a precipitate are made, as shown in Fig. 4.2.

**Fig. 4.2**

Table 4.1 describes the colour of each of the precipitates made.

Table 4.1

halogenoalkane added to $\text{AgNO}_3(\text{aq})$ in ethanol	colour of precipitate
Q	white
M	yellow

- (v) Deduce the structure of the straight-chain halogenoalkane **M**.

[2]

33. MAR 22/P22/Q3(e)(iii)

Context

- 3** The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

Context: part (e) (introduction)

- (e) HBr reacts with propene to form two bromoalkanes, $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and $(\text{CH}_3)_2\text{CHBr}$.

- (iii) The reaction of $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and NaOH is different depending on whether water or ethanol is used as a solvent.

Complete Table 3.2 to identify the organic and inorganic products of the reaction of $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and NaOH in each solvent.

Table 3.2

solvent	organic product(s)	inorganic product(s)
water		
ethanol		

[2]

34. M/J 21/P21/Q6(e)

Context

- 6** Propene, C_3H_6 , reacts with H_2O in the presence of an acid catalyst to form an alcohol with molecular formula $\text{C}_3\text{H}_8\text{O}$.

(e) 2-bromopropane reacts to form propene, hydrogen bromide and water under certain conditions.

- (i) Name this type of reaction.

[1]

- (ii) Describe the reagents and conditions needed to favour this reaction.

reagents

conditions

[2]

35. M/J 21/P22/Q3(c)

Context

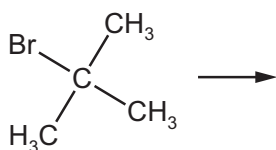
- 3** A large excess of 2-bromo-2-methylpropane is added to 0.0010 mol of $\text{NaOH}(\text{aq})$, which contains a few drops of phenolphthalein indicator. A stopwatch is started as soon as the substances are mixed. The time taken for the pink colour to disappear is recorded.

The experiment is repeated at different temperatures, keeping all concentrations and volumes of reagents constant.

temperature / $^{\circ}\text{C}$	time taken for pink colour to disappear/s
20	300
25	65
35	20

- (c) (i) Draw the mechanism to show the reaction of 2-bromo-2-methylpropane with $\text{OH}^-(\text{aq})$. Show the intermediate formed in this reaction.

Include all charges, partial charges, lone pairs and curly arrows as appropriate.



[3]

- (ii) Name the mechanism for this reaction.

[1]

36. M/J 21/P22/Q3(d)

Context

- 3** A large excess of 2-bromo-2-methylpropane is added to 0.0010 mol of NaOH(aq), which contains a few drops of phenolphthalein indicator. A stopwatch is started as soon as the substances are mixed. The time taken for the pink colour to disappear is recorded.

The experiment is repeated at different temperatures, keeping all concentrations and volumes of reagents constant.

temperature / °C	time taken for pink colour to disappear / s
20	300
25	65
35	20

- (d)** The original experiment is repeated at 25 °C with 2-chloro-2-methylpropane instead of 2-bromo-2-methylpropane. All other variables remain constant.

Predict the effect of using 2-chloro-2-methylpropane compared to 2-bromo-2-methylpropane on the time taken for the pink colour to disappear. Explain your answer.

[2]

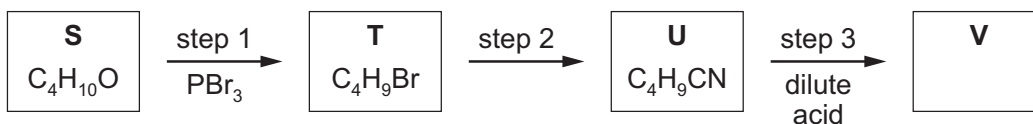
37. M/J 21/P22/Q5(b)(iii)

Context

- 5** **S** is a secondary alcohol with molecular formula C₄H₁₀O.

Context: part (b) (introduction)

- (b)** **S** is converted to **V** in a three-step reaction sequence.



In step 1, the secondary alcohol **S** reacts with PBr₃ to produce **T**, which has molecular formula C₄H₉Br.

- (iii)** State the reagent(s) and conditions for step 2.

[2]

38. M/J 20/P21/Q5(c)(ii)

- (ii)** Name an organic functional group which tends to react with a nucleophile in an S_N1 substitution mechanism.

[1]

39. M/J 20/P22/Q4(a)(i)-(ii)

Context: part (a) (introduction)

- 4 (a)** An unlabelled bottle contains a straight-chain halogenoalkane, **Q**. The molecular formula of **Q** is $C_5H_{11}X$, where **X** is a halogen; bromine, chlorine or iodine.

A test is carried out to identify the halogen present in **Q**.

A sample of **Q** is added to NaOH(aq) and warmed. Dilute nitric acid is then added followed by a few drops of aqueous silver nitrate. A cream precipitate is observed.

- (i) Suggest the identity of **X**.

[1]

- (ii) Write an ionic equation to describe the formation of the cream precipitate. Include state symbols.

[1]

40. M/J 20/P22/Q4(b)

Context: part (a) (introduction)

- 4 (a)** An unlabelled bottle contains a straight-chain halogenoalkane, **Q**. The molecular formula of **Q** is $C_5H_{11}X$, where **X** is a halogen; bromine, chlorine or iodine.

A test is carried out to identify the halogen present in **Q**.

A sample of **Q** is added to NaOH(aq) and warmed. Dilute nitric acid is then added followed by a few drops of aqueous silver nitrate. A cream precipitate is observed.

- (b) The reaction of **Q** with NaOH(aq) tends to proceed via an S_N2 mechanism.

- (i) Suggest the structural formula of the straight-chain halogenoalkane **Q**.

[1]

- (ii) Explain why the reaction tends to proceed via an S_N2 mechanism rather than an S_N1 mechanism.

[2]

41. M/J 20/P22/Q4(c)(i)-(ii)

Context: part (c) (introduction)

- (c) Two different halogenoalkanes, **P** and **R**, both with the molecular formula C_4H_9Cl , are separately dissolved in ethanol and heated under reflux with sodium hydroxide.

The major organic product of each of these reactions is methylpropene.

- (i) Name the type of reaction occurring.

[1]

- (ii) Write an equation, using molecular formulae, to represent the reaction occurring.

[1]

42. MAR 20/P22/Q2(e)(iv)

Context

- 2** The Group 17 elements, chlorine, bromine and iodine, are non-metals that show trends in their physical and chemical properties.

Context: part (e) (introduction)

- (e)** Lucas's reagent is a mixture of HCl and ZnCl_2 . Primary, secondary and tertiary alcohols can be distinguished by their reaction with Lucas's reagent.

Alcohols react with the HCl in Lucas's reagent to form halogenoalkanes.

ZnCl_2 acts as a homogeneous catalyst for these reactions.

- (iv)** Tertiary alcohols tend to react with Lucas's reagent using the same mechanism as in their reaction with HCl .

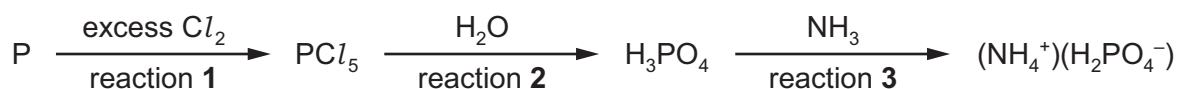
Suggest the type of reaction shown by tertiary alcohols with Lucas's reagent.

[1]

43. O/N 19/P22/Q3(c)(iv)

Context

- 3** A series of reactions for phosphorus and its compounds is shown.



Context: part (c) (introduction)

- (c)** PCl_5 can be used to convert alcohols to halogenoalkanes.

- (iv)** The rate of the hydrolysis reaction of halogenoalkanes with NaOH(aq) is dependent on the halogen that is bonded to carbon.

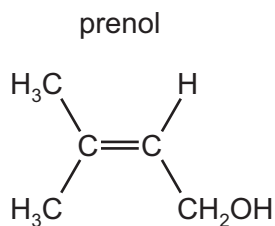
State and explain the order of reactivity when NaOH(aq) reacts separately with $\text{C}_2\text{H}_5\text{Cl}$, $\text{C}_2\text{H}_5\text{Br}$ and $\text{C}_2\text{H}_5\text{I}$.

[2]

44. O/N 19/P22/Q4(a)

Context

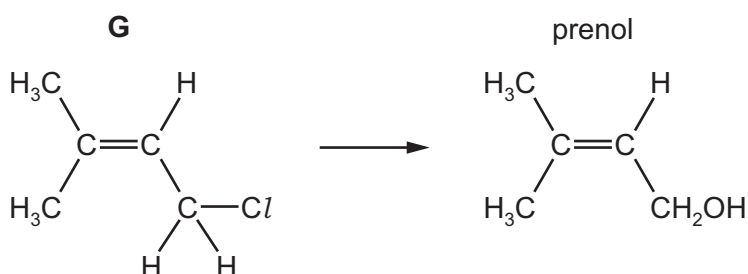
- 4** Prenol is a naturally occurring organic molecule found in many fruits. It contains both an alkene and an alcohol functional group.



- (a)** Prenol can be formed by the reaction of **G** with NaOH(aq).

Complete the diagram to show the mechanism of the reaction between **G** and NaOH(aq) to form prenol.

Include all relevant charges, partial charges, lone pairs and curly arrows.

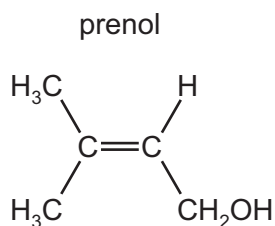


[2]

45. O/N 19/P22/Q4(e)(ii)

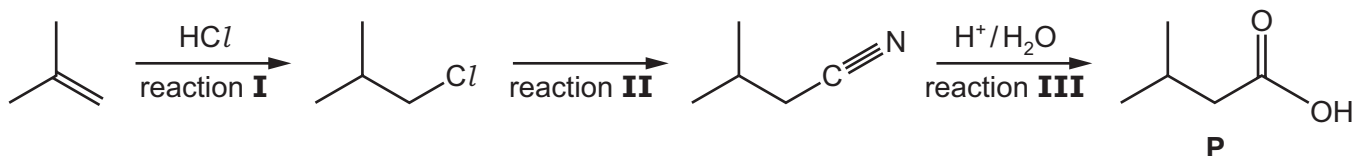
Context

- 4** Prenol is a naturally occurring organic molecule found in many fruits. It contains both an alkene and an alcohol functional group.



Context: part (e) (introduction)

- (e)** **P** can be produced as shown.



- (ii)** State the reagent(s) needed for reaction **II**.

[1]

46. M/J 19/P23/Q5(a)(ii)-(iv)

Context

5 Halogenoalkanes react with a number of different reagents in nucleophilic substitution reactions.

Context: part (a) (introduction)

(a) A sample of potassium cyanide dissolved in ethanol is added to a sample of 1-bromobutane, $\text{CH}_3(\text{CH}_2)_3\text{Br}$, and heated under reflux. A nucleophilic substitution reaction occurs and compound **A** is formed.

(ii) What is meant by the term *nucleophile*?

[1]

(iii) Identify the nucleophile in this reaction.

[1]

(iv) Explain why this reaction is described as a substitution reaction.

[1]

47. M/J 19/P23/Q5(c)

Context

5 Halogenoalkanes react with a number of different reagents in nucleophilic substitution reactions.

(c) Equal amounts of three different halogenoalkanes are added to three separate test-tubes. An equal amount of aqueous silver nitrate and ethanol is added to each test-tube. The time taken for a precipitate to form is recorded for each halogenoalkane.

halogenoalkane	time taken for precipitate to form/s
$(\text{CH}_3)_3\text{CCl}$	460
$(\text{CH}_3)_3\text{CBr}$	190
$(\text{CH}_3)_3\text{CI}$	40

(i) Describe and explain the trend in reactivity of the different halogenoalkanes shown in this experiment.

[2]

(ii) All three halogenoalkanes tend to react via the $\text{S}_{\text{N}}1$ mechanism.

Explain why the $\text{S}_{\text{N}}1$ mechanism is favoured.

[3]

(iii) Identify a halogenoalkane which tends to react with an aqueous solution of silver nitrate and ethanol via the $\text{S}_{\text{N}}2$ mechanism.

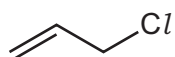
[1]

48. MAR 19/P22/Q4(c)(iii)

Context

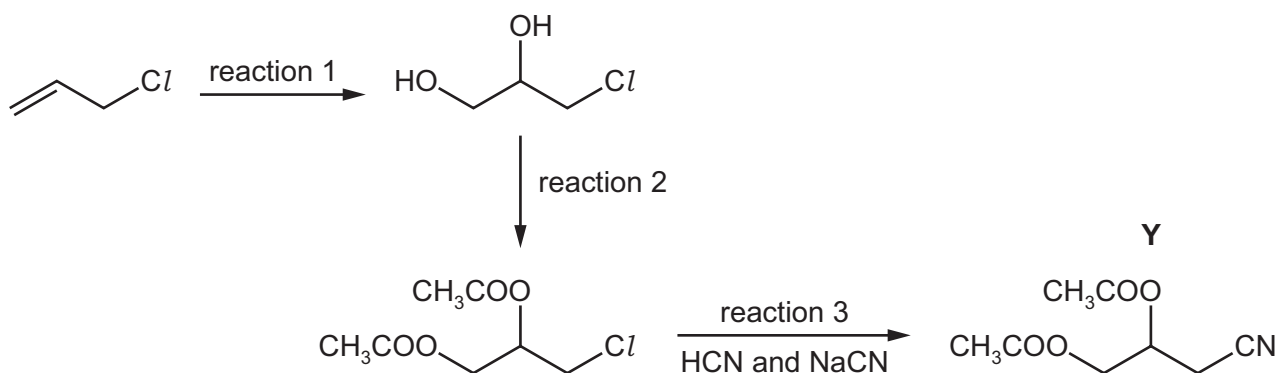
- 4** Allyl chloride is an important chemical used in the manufacture of plastics, pharmaceuticals and pesticides.

allyl chloride



Context: part (c) (introduction)

(c) A series of reactions starting from allyl chloride is shown.



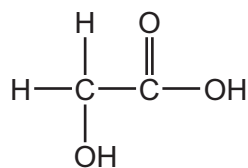
- (iii)** State the name of the mechanism that occurs in reaction 3.

[1]

49. O/N 18/P21/Q4(b)(v)

Context

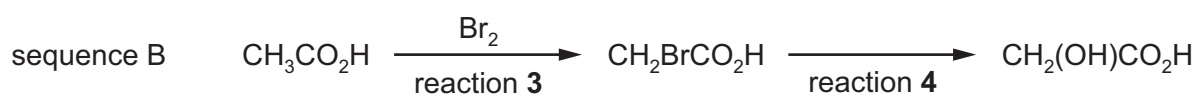
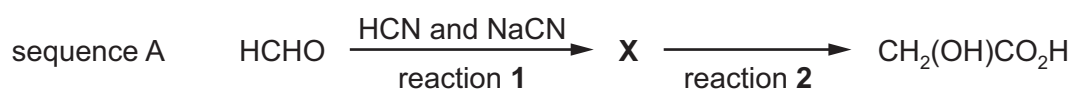
4 The structure of glycolic acid is shown.



glycolic acid

Context: part (b) (introduction)

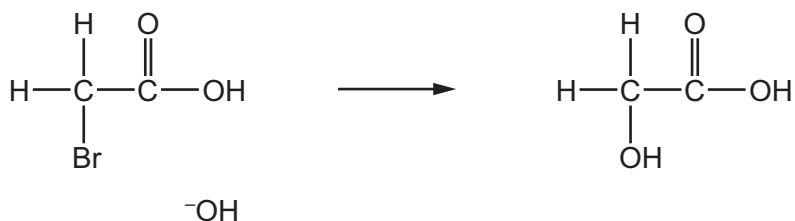
(b) Two reaction sequences to make glycolic acid are shown.



(v) Reaction **4** occurs via an $\text{S}_{\text{N}}2$ mechanism.

Complete the diagram for the mechanism for reaction **4**.

Include all relevant charges, partial charges, curly arrows and lone pairs.

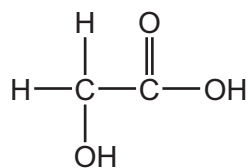


[2]

50. O/N 18/P23/Q4(b)(v)

Context

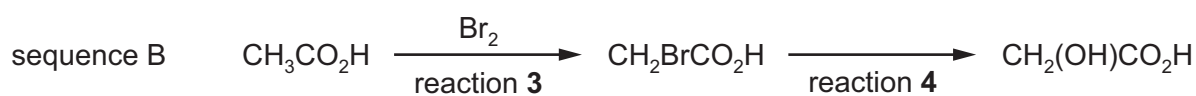
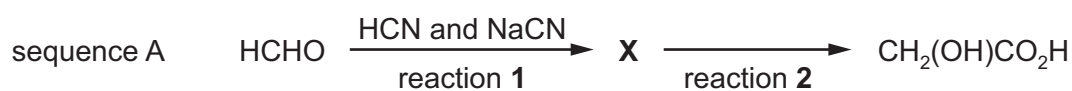
4 The structure of glycolic acid is shown.



glycolic acid

Context: part (b) (introduction)

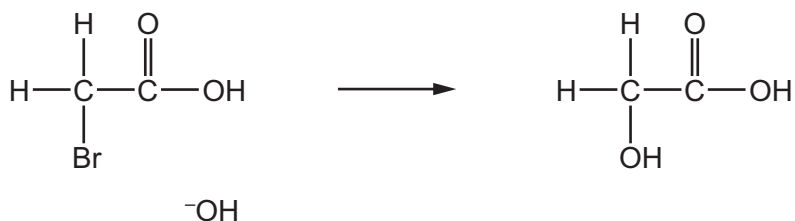
(b) Two reaction sequences to make glycolic acid are shown.



(v) Reaction **4** occurs via an $\text{S}_{\text{N}}2$ mechanism.

Complete the diagram for the mechanism for reaction **4**.

Include all relevant charges, partial charges, curly arrows and lone pairs.



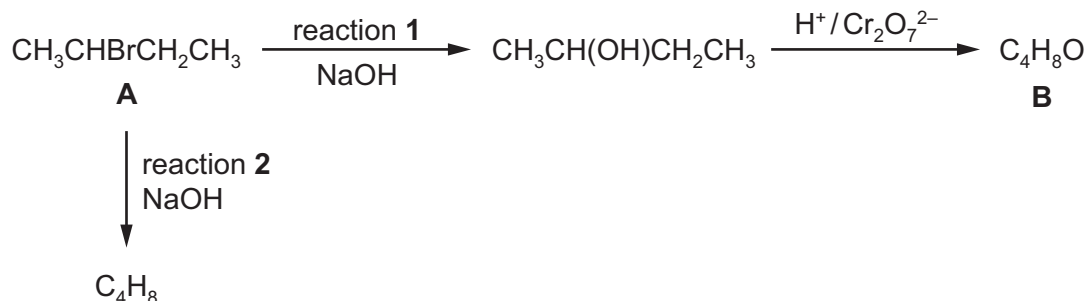
[2]

51. M/J 18/P23/Q4(c)

Context

4 **A** is $\text{CH}_3\text{CHBrCH}_2\text{CH}_3$.

Context: part (a) (introduction)

(a) Some reactions of **A** are shown.**(c)** Reaction **1** occurs by two different mechanisms at the same time.These mechanisms are referred to as $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$.**(i)** State what the letters 'S' and 'N' represent in the abbreviation $\text{S}_{\text{N}}1$.

S

N

[1]

(ii) Complete the $\text{S}_{\text{N}}1$ mechanism for reaction **1**.

Include the structure of the intermediate and all necessary charges, dipoles, lone pairs and curly arrows.



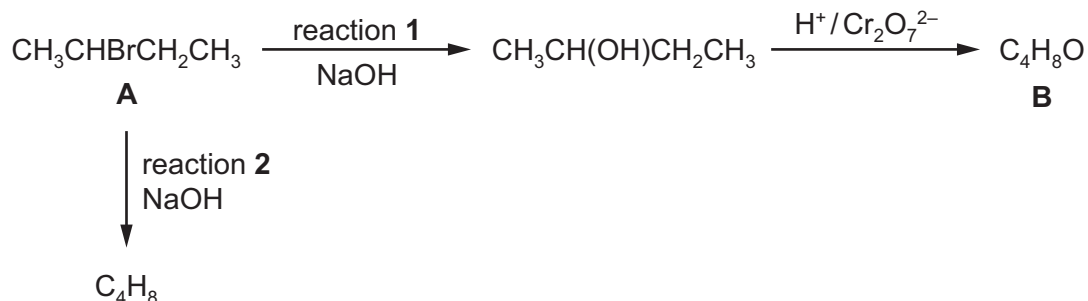
[3]

52. M/J 18/P23/Q4(d)

Context

4 **A** is $\text{CH}_3\text{CHBrCH}_2\text{CH}_3$.

Context: part (a) (introduction)

(a) Some reactions of **A** are shown.**(d)** The $\text{S}_{\text{N}}1$ mechanism for reaction 1 is repeated using $\text{CH}_3\text{CHClCH}_2\text{CH}_3$ or $\text{CH}_3\text{CHICH}_2\text{CH}_3$ in place of the $\text{CH}_3\text{CHBrCH}_2\text{CH}_3$.State and explain how the rates of these two reactions will compare with the rate of the original reaction using $\text{CH}_3\text{CHBrCH}_2\text{CH}_3$.

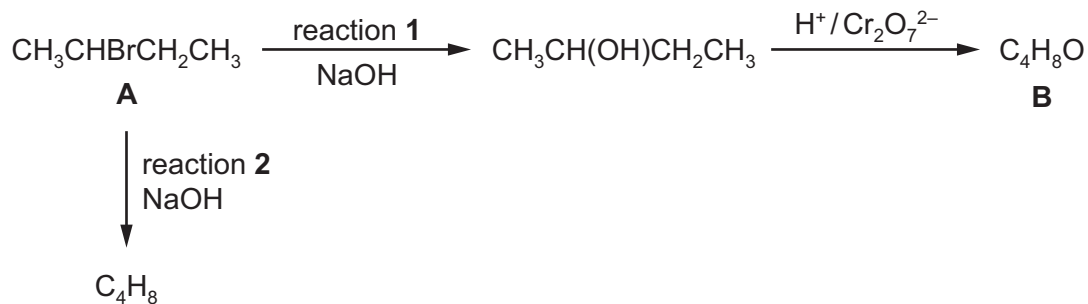
[3]

53. M/J 18/P23/Q4(e)

Context

4 **A** is $\text{CH}_3\text{CHBrCH}_2\text{CH}_3$.

Context: part (a) (introduction)

(a) Some reactions of **A** are shown.**(e)** Reaction 2 uses the same reagent as reaction 1, but under different conditions.State **two** differences in the conditions needed to ensure that reaction 2 is more likely to take place than reaction 1 when this reagent is added.

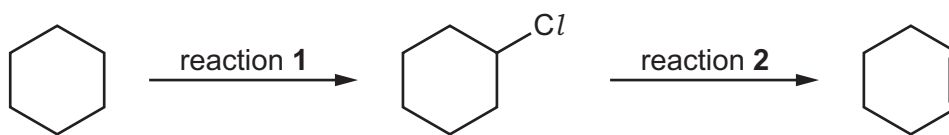
[2]

54. MAR 18/P22/Q4(b)

Context

- 4** Cyclohexane is a colourless liquid used in industry to produce synthetic fibres.

A reaction scheme involving cyclohexane is shown.



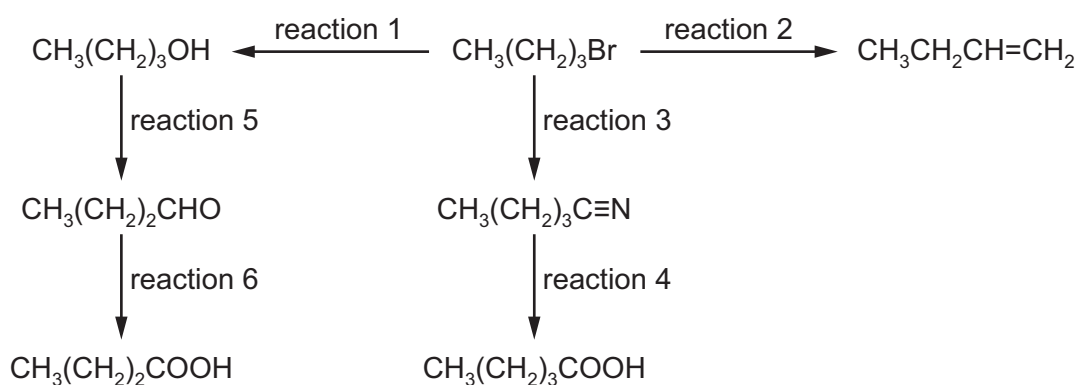
- (b)** Name the type of reaction that occurs in reaction 2.

[1]

55. O/N 17/P21/Q3(a)

Context

- 3** Some reactions based on 1-bromobutane, $\text{CH}_3(\text{CH}_2)_3\text{Br}$, are shown.



- (a)** For each of the reactions state the reagent(s), the particular conditions required, if any, and the type of reaction.

For the type of reaction choose from the list.

Each type may be used once, more than once or not at all.

Each reaction may be described by more than one type.

elimination hydrolysis substitution

oxidation addition condensation

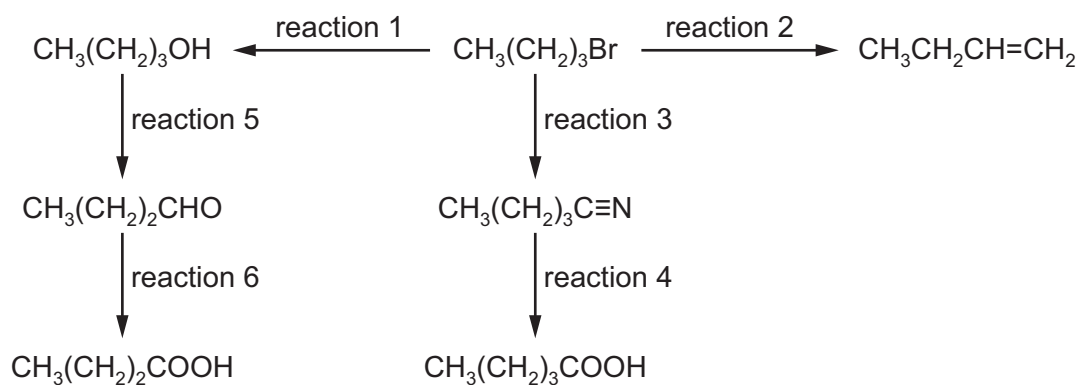
reaction	reagent(s) and conditions	type(s) of reaction
1		
2		
3		
4		
5		
6		

[6]

56. O/N 17/P21/Q3(b)

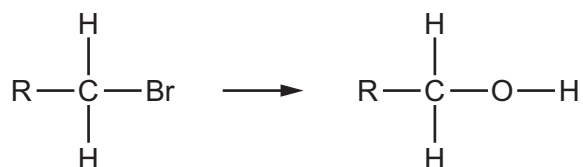
Context

3 Some reactions based on 1-bromobutane, $\text{CH}_3(\text{CH}_2)_3\text{Br}$, are shown.



(b) Complete the diagram to show the $\text{S}_{\text{N}}2$ mechanism of reaction 1. R represents the $\text{CH}_3(\text{CH}_2)_2$ group.

Include all necessary charges, dipoles, lone pairs and curly arrows.

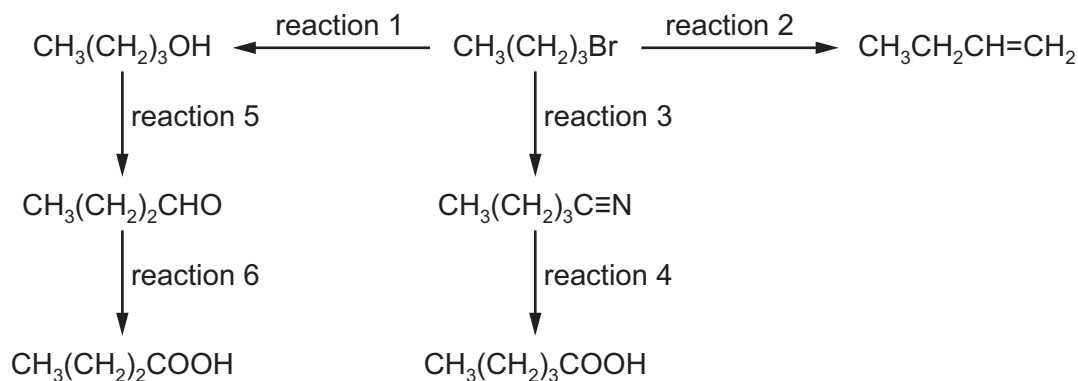


[2]

57. O/N 17/P21/Q3(c)(ii)

Context

3 Some reactions based on 1-bromobutane, $\text{CH}_3(\text{CH}_2)_3\text{Br}$, are shown.



Context: part (c) (introduction)

(c) 2-bromo-2-methylpropane is a tertiary halogenoalkane that is a structural isomer of 1-bromobutane.

(ii) 2-bromo-2-methylpropane is treated with the same reagents as in reaction 1. Methylpropan-2-ol is formed.

Identify the mechanism for this reaction.

Explain why this reaction proceeds via a different mechanism from that of reaction 1.

mechanism

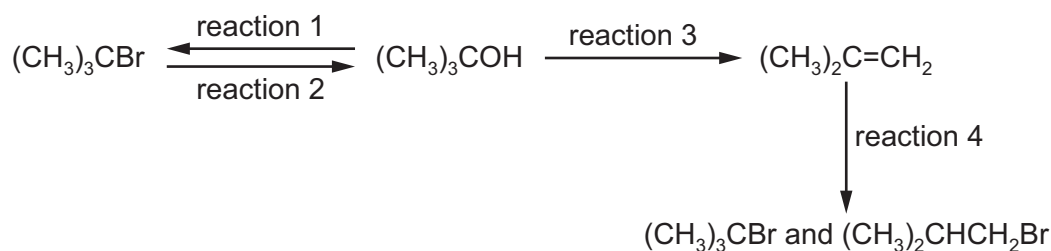
explanation

[3]

58. O/N 17/P22/Q4(b)

Context

4 Some reactions are shown, based on methylpropan-2-ol, $(\text{CH}_3)_3\text{COH}$.



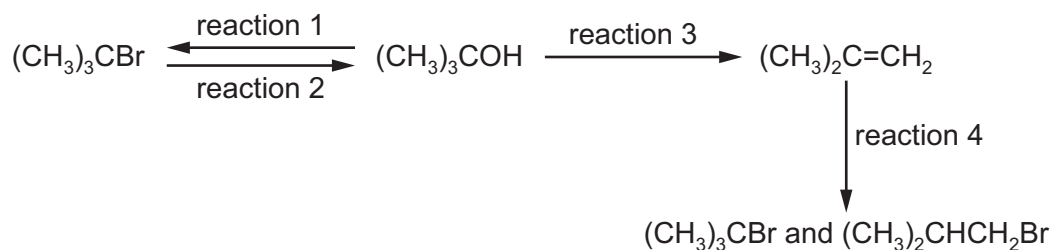
(b) Draw a diagram to show the $\text{S}_{\text{N}}1$ mechanism of reaction 2. Include all necessary charges, dipoles, lone pairs and curly arrows.

[3]

59. O/N 17/P22/Q4(c)(ii)

Context

4 Some reactions are shown, based on methylpropan-2-ol, $(\text{CH}_3)_3\text{COH}$.



Context: part (c) (introduction)

(c) 1-bromobutane is a structural isomer of the product of reaction 1.

(ii) 1-bromobutane is treated with the same reagents as in reaction 2. Butan-1-ol is formed.

Identify the mechanism of this reaction.

Explain why this reaction proceeds via a different mechanism from that of reaction 2.

mechanism

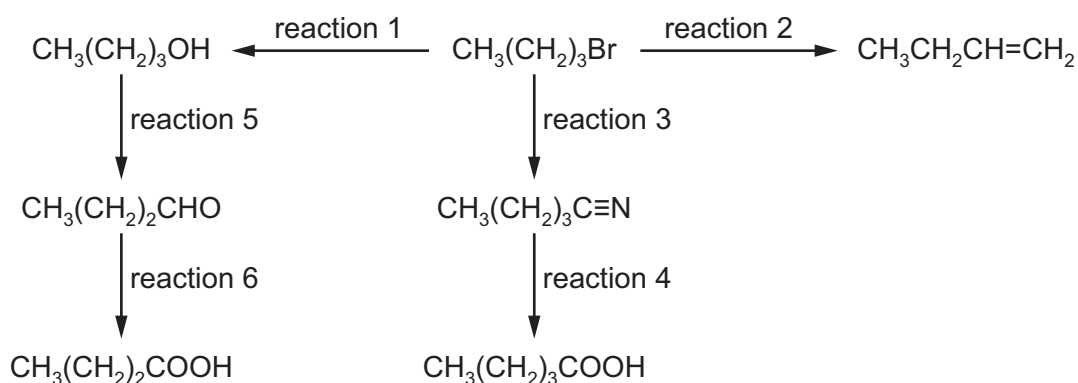
explanation

[3]

60. O/N 17/P23/Q3(a)

Context

3 Some reactions based on 1-bromobutane, $\text{CH}_3(\text{CH}_2)_3\text{Br}$, are shown.



(a) For each of the reactions state the reagent(s), the particular conditions required, if any, and the type of reaction.

For the type of reaction choose from the list.

Each type may be used once, more than once or not at all.

Each reaction may be described by more than one type.

elimination

hydrolysis

substitution

oxidation

addition

condensation

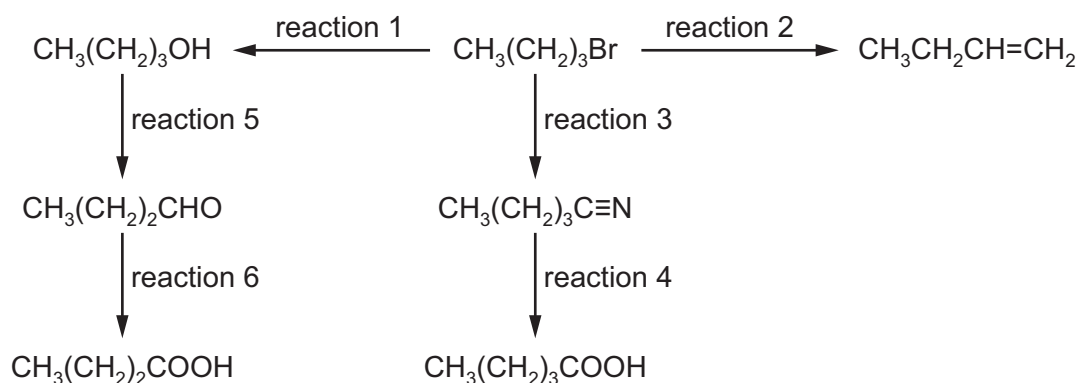
reaction	reagent(s) and conditions	type(s) of reaction
1		
2		
3		
4		
5		
6		

[6]

61. O/N 17/P23/Q3(b)

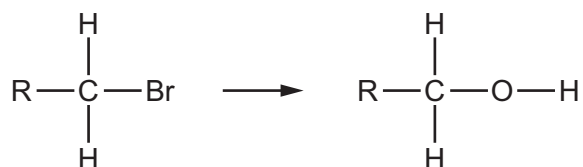
Context

3 Some reactions based on 1-bromobutane, $\text{CH}_3(\text{CH}_2)_3\text{Br}$, are shown.



(b) Complete the diagram to show the $\text{S}_{\text{N}}2$ mechanism of reaction 1. R represents the $\text{CH}_3(\text{CH}_2)_2$ group.

Include all necessary charges, dipoles, lone pairs and curly arrows.

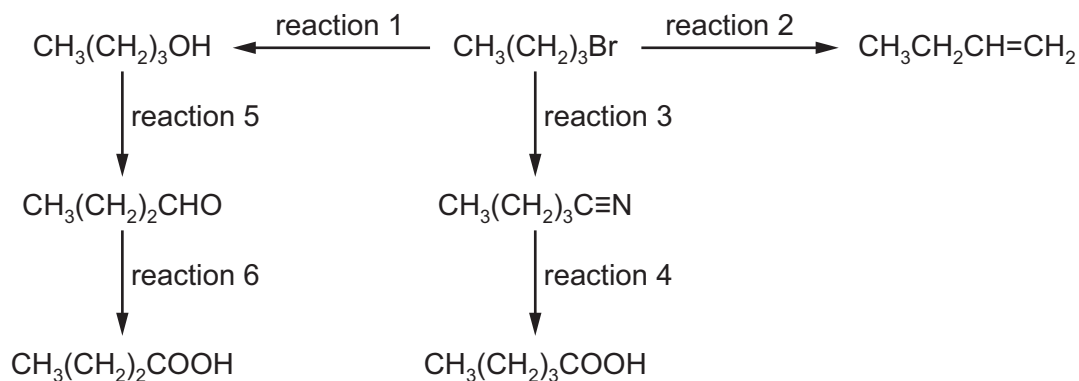


[2]

62. O/N 17/P23/Q3(c)(ii)

Context

3 Some reactions based on 1-bromobutane, $\text{CH}_3(\text{CH}_2)_3\text{Br}$, are shown.



Context: part (c) (introduction)

(c) 2-bromo-2-methylpropane is a tertiary halogenoalkane that is a structural isomer of 1-bromobutane.

(ii) 2-bromo-2-methylpropane is treated with the same reagents as in reaction 1. Methylpropan-2-ol is formed.

Identify the mechanism for this reaction.

Explain why this reaction proceeds via a different mechanism from that of reaction 1.

mechanism

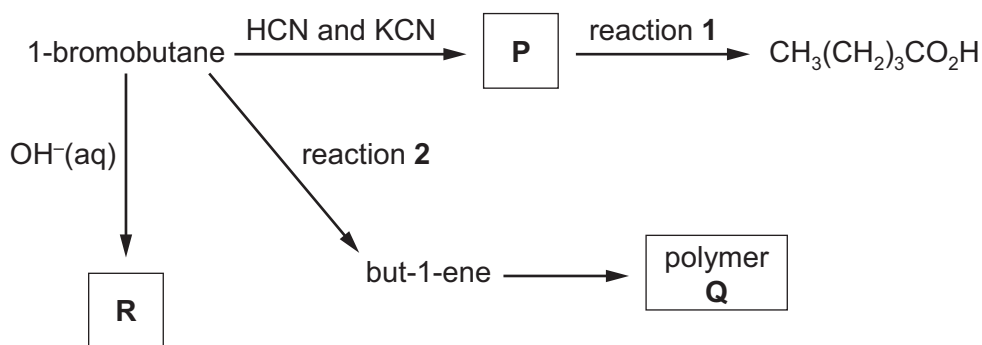
explanation

[3]

63. MAR 17/P22/Q3(a)(i)

Context: part (a) (introduction)

3 (a) A series of reactions starting from 1-bromobutane is shown.

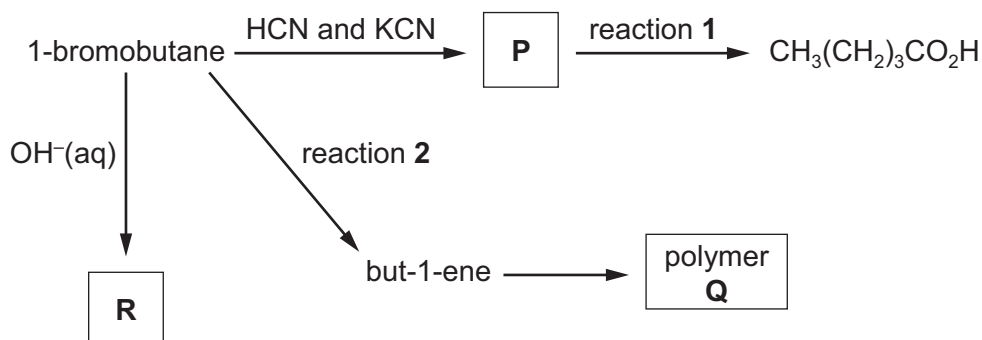
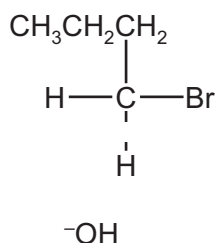


(i) Draw the **displayed** formula of compound **P**.

[1]

64. MAR 17/P22/Q3(b)

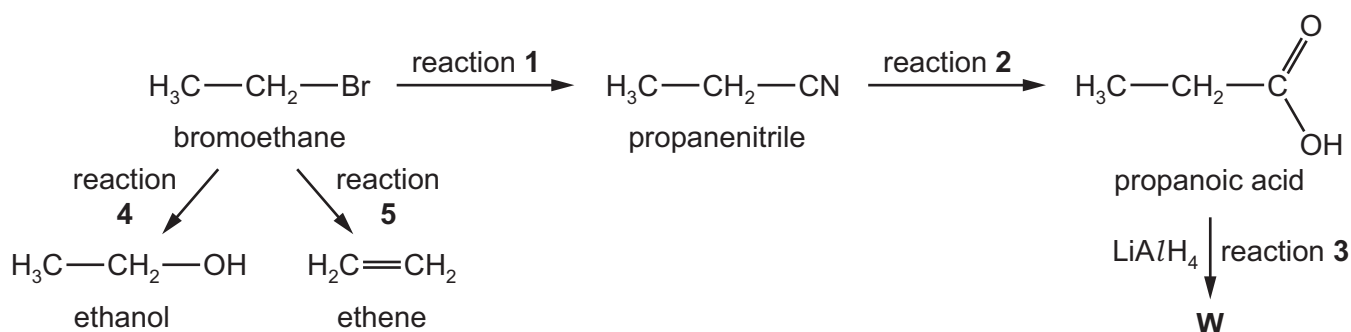
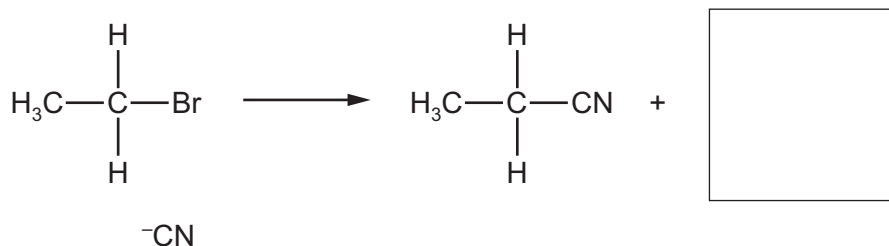
Context: part (a) (introduction)

3 (a) A series of reactions starting from 1-bromobutane is shown.**(b)** Complete the reaction scheme to show the mechanism of the reaction of 1-bromobutane with $\text{OH}^-(\text{aq})$ to produce **R**.Include all necessary charges, dipoles, lone pairs and curly arrows and the structure of **R**.

[3]

65. M/J 16/P21/Q5(a)

Context

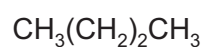
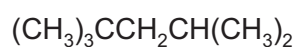
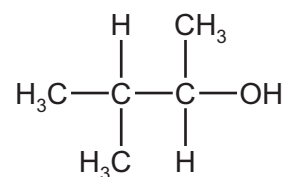
5 A reaction sequence is shown.**(a)** Complete the diagram to show the mechanism of reaction 1. Include all necessary charges, partial charges, lone pairs and curly arrows.

[2]

66. MAR 16/P22/Q4(e)

Context

- 4** The following compounds were all found to be components of a sample of petrol.

**G****H****J**

- (e)** Compound **J** can be produced from 2-chloro-3-methylbutane, $\text{C}_5\text{H}_{11}\text{Cl}$.

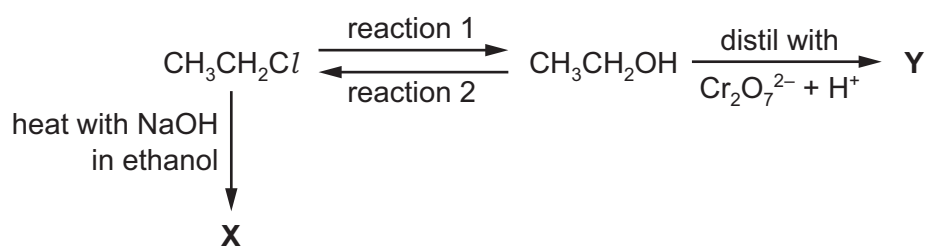
Give the reagent(s) and conditions for this reaction.

[1]

67. O/N 15/P21/Q4(a)(i)

Context

- 4** Some reactions involving ethanol are shown.



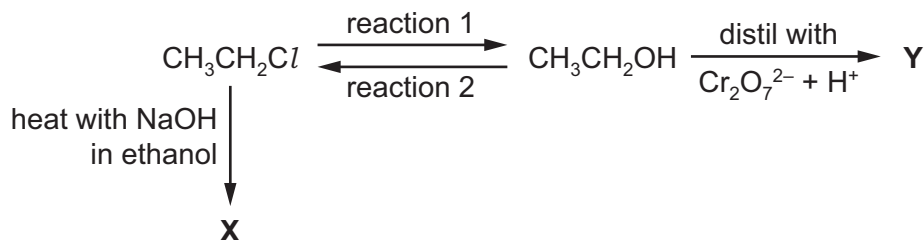
- (a) (i)** Give an equation for reaction 2 including the reagent needed for the conversion.

[2]

68. O/N 15/P21/Q4(b)

Context

4 Some reactions involving ethanol are shown.



(b) (i) Identify the organic product **X**.

[1]

(ii) Nitric acid is added to the products of reaction of $\text{CH}_3\text{CH}_2\text{Cl}$ with NaOH in ethanol. Silver nitrate solution is then added to this mixture.

State what you would observe.

[1]

(iii) Write an ionic equation, including state symbols, for the reaction responsible for the observation in **(ii)**.

[1]

69. O/N 15/P22/Q4(a)(i)

Context

4 Halogenoalkanes are useful intermediates in the synthesis of a wide variety of compounds.

Context: part (a) (introduction)

(a) 2-bromobutane reacts in two different ways with sodium hydroxide depending on the conditions.

When warmed with aqueous sodium hydroxide, 2-bromobutane produces an alcohol that exists as a pair of optical isomers.

(i) Give the name of the mechanism of the reaction between 2-bromobutane and aqueous sodium hydroxide.

[1]

70. O/N 15/P22/Q4(a)(iv)

Context

4 Halogenoalkanes are useful intermediates in the synthesis of a wide variety of compounds.

Context: part (a) (introduction)

(a) 2-bromobutane reacts in two different ways with sodium hydroxide depending on the conditions.

When warmed with aqueous sodium hydroxide, 2-bromobutane produces an alcohol that exists as a pair of optical isomers.

(iv) Give the name of the mechanism of the reaction between 2-bromobutane and ethanolic sodium hydroxide.

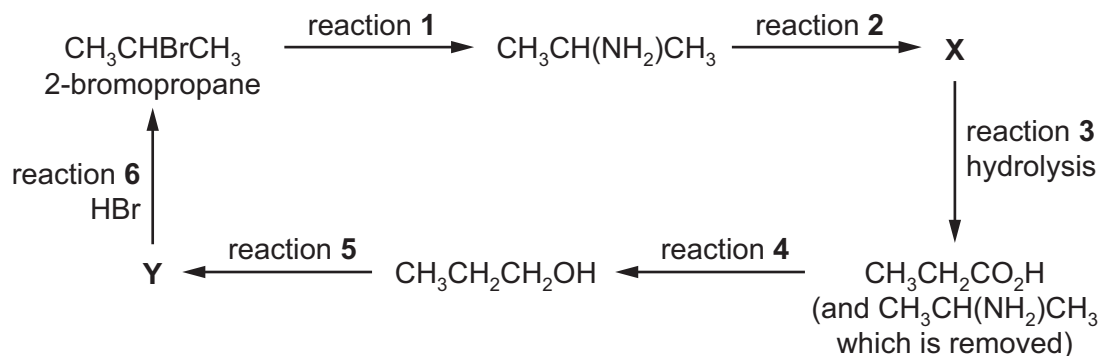
[1]

71. O/N 15/P22/Q4(b)(ii)-(iii)

Context

4 Halogenoalkanes are useful intermediates in the synthesis of a wide variety of compounds.

Context: part (b) (introduction)

(b) Some reactions involving 2-bromopropane are shown.**(ii)** State the reagent needed for reaction 2.

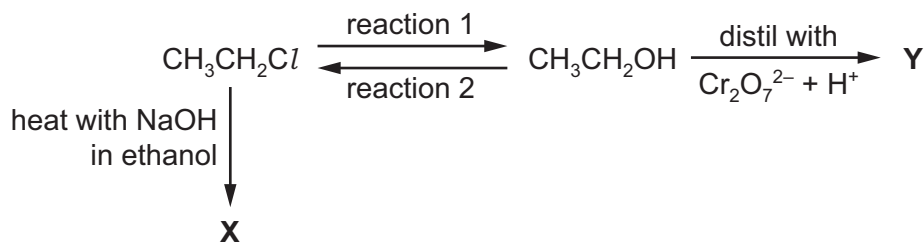
[1]

(iii) Give the structural formula of **X**.

[1]

72. O/N 15/P23/Q4(a)(i)

Context

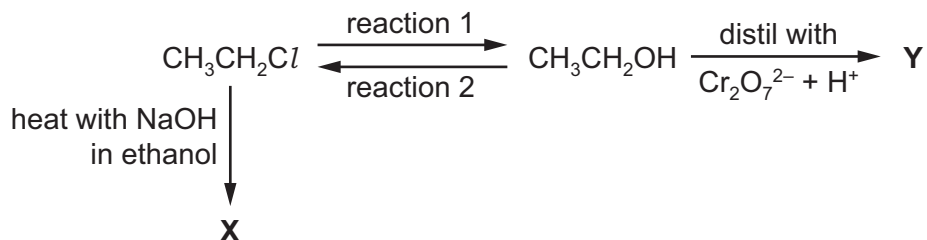
4 Some reactions involving ethanol are shown.**(a) (i)** Give an equation for reaction 2 including the reagent needed for the conversion.

[2]

73. O/N 15/P23/Q4(b)

Context

4 Some reactions involving ethanol are shown.



(b) (i) Identify the organic product **X**.

[1]

(ii) Nitric acid is added to the products of reaction of $\text{CH}_3\text{CH}_2\text{Cl}$ with NaOH in ethanol. Silver nitrate solution is then added to this mixture.

State what you would observe.

[1]

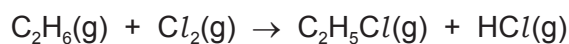
(iii) Write an ionic equation, including state symbols, for the reaction responsible for the observation in **(ii)**.

[1]

74. M/J 15/P22/Q3(b)(ii)

Context

3 Ethane reacts with chlorine to form chloroethane.



Context: part (b) (introduction)

(b) Chloroethane can be converted back into ethane by a two-stage process via an intermediate compound, **X**.



(ii) Suggest the reagent and conditions needed for reaction 1.

[2]

Answers Section

1. M/J 26/P21/Q5(e)

5(e)	elimination	1
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2. MAR 26/P22/Q3(a)

3(a)	(organic molecule where) carbon (of C—Hal) is bonded to two other C atoms	1
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3. MAR 26/P22/Q3(d)

3(d)	M1 fastest $D > C > B$ slowest M2 carbon—halogen gets weaker down the group (so breaks faster)	2
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4. MAR 26/P22/Q4(c)

4(c)(i)	M1 curly arrow from lone pair on C of CN^- to δ^+C of $C-Cl$ M2 correct dipole of $\delta^+C-Cl^{\delta-}$ AND curly arrow from bond to Cl	2
4(c)(ii)	CN^- / cyanide	1
4(c)(iii)	nucleophilic substitution / $S_N(2)$	1

5. O/N 25/P22/Q4(b)

4(b)	sodium hydroxide AND water	1
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6. O/N 25/P24/Q3(b)(ii)-(iii)

3(b)(ii)	M1 correct dipole on $\delta^+C-Br^{\delta-}$ AND curly arrow from $C-Br$ bond to Br  M2 intermediate M3 curly arrow from lone pair of $(CH_3)_2CHO^-$ to C^+	3
3(b)(iii)	M1 rate would decrease M2 $C-Br$ is weaker (than $C-Cl$, so breaks more easily)	2

7. O/N 25/P24/Q4(b)(i)

4(b)(i)	KCN in ethanol (and heat)	1
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Answer for context part (a)(v)

4(a)(v)	M1 (reaction proceeds via a) more stable intermediate / carbocation M2 because of greater positive inductive effect of more alkyl groups	2
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8. M/J 25/P22/Q4(b)(ii)

4(b)(ii)	KCN / potassium cyanide in ethanol / C_2H_5OH AND heat	1
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9. M/J 25/P23/Q6(c)(i)

6(c)(i)	KCN/potassium cyanide in ethanol AND heat	1
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10. MAR 25/P22/Q1(c)(ii)

1(c)(ii)	ethanol (solvent) AND heat	1
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11. O/N 24/P21/Q2(c)(iv)

2(c)(iv)	M1 white precipitate/white ppt forms	1
	M2 CHCl_3 is hydrolysed / chloride ions are released OR Cl^- / chloride ions which are released react with Ag^+ OR silver chloride / AgCl is formed	1

12. O/N 24/P23/Q2(c)(iv)

2(c)(iv)	M1 white precipitate/white ppt forms	1
	M2 CHCl_3 is hydrolysed / chloride ions are released OR Cl^- / chloride ions which are released react with Ag^+ OR silver chloride / AgCl is formed	1

13. M/J 24/P22/Q4(b)

4(b)	M1 correct dipole ($\delta^+\text{C}-\text{Br}^{\delta-}$) AND curly arrow from bond of $\text{C}-\text{Br}$ to Br (or just beyond) M2 correct intermediate ALLOW R for $-(\text{CH}_2)_5\text{CH}_3$ M3 curly arrow from lone pair on OH^- to the C of intermediate (or suitable position where $\text{C}-\text{OH}$ bond is made) <i>If $\text{S}_{\text{N}}2$ mechanism shown:</i> Apply $\text{S}_{\text{N}}2$ mechanism mark scheme only if mechanism shows OH^- / NaOH species attacking the reactant and no OH M1 correct dipole ($\delta^+\text{C}-\text{Br}^{\delta-}$) AND curly arrow from bond of $\text{C}-\text{Br}$ to Br of reactant (or just beyond) NOT M2 M3 curly arrow from lone pair on $:\text{OH}^-$ to the appropriate C of the reactant	3
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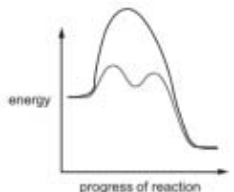
14. M/J 24/P22/Q4(d)

4(d)(i)	elimination	1
4(d)(ii)	NaOH / sodium hydroxide in ethanol	1

15. M/J 24/P23/Q4(c)

4(c)(i)	nucleophile	1
4(c)(ii)	OH^- (more reactive because) negatively charged species more attracted to halogenoalkane and / or $\text{C}^{\delta+}$ and / or intermediate and / or C^+ OR NH_3 (more reactive because) N electrons less tightly held OR N is more able to donate (lp) electrons (to make the covalent bond) OR N is less electronegative ORA	1

16. M/J 24/P23/Q4(d)

4(d)	 M1 same start energy and finishing energy AND single hump M2 transition state / activation energy shown in answer is higher than both transition state energies shown in reaction profile (as the reaction is slower)	2
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17. M/J 24/P23/Q4(e)(i)-(ii)

4(e)(i)	M1 dipole and correct arrow on C—Br of 2-bromobutane M2 lone pair on NH₃ attacking C(+) of the intermediate drawn M3 curly arrow showing bond between N—H breaking and pointing towards the N	3
4(e)(ii)	NH ₄ Br / ammonium bromide	1

18. M/J 24/P23/Q4(f)

4(f)(i)		1
4(f)(ii)	M1 2-bromo-2-methylpropane AND more stable carbocation / C⁺ M2 alkyl groups have an inductive effect (of alkyl groups) AND 2-bromo-2-methylpropane has more / most alkyl groups attached (to C⁺) / carbocation contains more alkyl groups	2

19. M/J 24/P23/Q5(b)(i)

5(b)(i)	step 1Br₂ (in the dark) step 2NaOH(aq) + heat	2
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Answer for context part (a)

5(a)	cold dilute KMnO ₄ OR cold dilute potassium manganate(VII)	1
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20. MAR 24/P22/Q2(d)(ii)

2(d)(ii)	NaOH in ethanol AND heat	1
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21. MAR 24/P22/Q4(b)(ii)

4(b)(ii)						1
	geometrical	✓	(×)	(×)	(×)	
	optical	(×)	✓	✓	✓	
CH ₃ CHBrCH ₂ CH ₃ nucleophilic substitution						

22. M/J 23/P21/Q3(e)

3(e)	CHCl ₃ + 2HF → CHClF ₂ + 2HCl	1
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23. M/J 23/P22/Q3(a)(i)

3(a)(i)	KCN in ethanol + heat	1
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24. MAR 23/P22/Q4(a)(i)

4(a)(i)	NaOH in alcohol / ethanol AND heat (under reflux)	1
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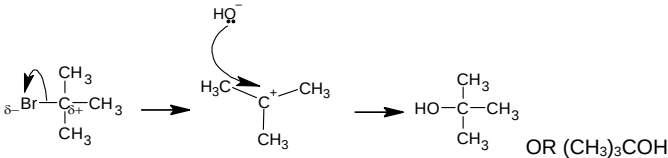
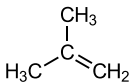
25. MAR 23/P22/Q4(b)(i)

4(b)(i)	cream(-coloured) / off-white precipitate (forms)	1
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26. MAR 23/P22/Q4(c)(i)

4(c)(i)	substitution	1
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27. O/N 22/P22/Q1(e)

1(e)(i)	 M1 correct dipole in on $\delta^+C-Br^{\delta-}$ (in tBuBr) AND curly arrow from $C-Br$ to Br M2 curly arrow from electron pair on O of $:OH^-$ to C^+ in carbocation M3 tBuOH product	1
1(e)(ii)	The $C-Br$ carbon is bonded to three other carbon atoms <i>owtte</i>	1
1(e)(iii)		1

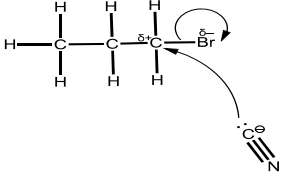
28. M/J 22/P22/Q5(a)

5(a)	reaction	reagent and conditions	type of reaction	6
	1	NaOH(aq) [M1]	substitution	
	2	NH ₃ [M2] in ethanol AND heat under pressure [M3]	substitution	
	3	NaOH in ethanol AND heat [M4]	elimination	
	M5 and M6 types of reaction [3 correct types = 2 marks and 2 correct types = 1 mark] All reagents can be identified as (correct) formula or in words			

29. M/J 22/P22/Q5(b)

5(b)	M1 C–I (covalent bond) is weaker / or a M2 lower activation energy / lower E_a (with 2-iodopropane) OR M2 explain in terms of the S_N1 mechanism (that is dominant / preferred / occurring with 2-iodopropane is fast(er)) (identical) carbocation is made (more) quickly (with 2-iodopropane) OR low / less energy is required to make the (same) carbocation / intermediate (from 2-iodopropane) OR M2 iodine / I (of C–I) has weaker attraction of nucleus to bonding / shared pair (of electrons) due to any one of: • more / high shielding (of electrons from inner shells) • more / 5 electron shells in iodine • C–I greater distance from nuclei (to bonding pair of electrons)	2
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30. M/J 22/P22/Q5(c)(i)

5(c)(i)	 M1 dipole $C^{\delta+}-Br^{\delta-}$ AND curly arrow from C–Br bond to $Br^{\delta-}$ M2 lone pair on C of ^-CN AND curly arrow from lone pair to C (on C-1 of bromopropane)	2
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31. M/J 22/P23/Q4(b)(i)-(iii)

4(b)(i)	M1 hydroxy(l) M2 substitution	2
4(b)(ii)	$I^- + Ag^+ \rightarrow AgI$	1
4(b)(iii)	M AND C–I (bond) is weak(er)	1

32. M/J 22/P23/Q4(b)(v)

4(b)(v)	$CH_3CH_2CHICH_3$ M1 correct identification of I on 4C molecule M2 unambiguous structure of M – unbranched with 4C and I on 2C	2
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33. MAR 22/P22/Q3(e)(iii)

3(e)(iii)	M1: (water solvent) $CH_3CH_2CH_2OH$ AND NaBr / sodium bromide / propan-1-ol M2: (ethanol solvent) CH_3CHCH_2 AND H_2O / water / propene AND NaBr / sodium bromide	2
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34. M/J 21/P21/Q6(e)

6(e)(i)	elimination	1
6(e)(ii)	M1 NaOH / KOH M2 ethanolic solution / ethanol / alcohol + heat	1

35. M/J 21/P22/Q3(c)

3(c)(i)	M1 correct dipole on haloalkane AND arrow from bond to Br or just beyond M2 correct intermediate M3 arrow from lone pair on <u>O</u> of OH⁻ / -OH to central C of their intermediate shown in M2 OR arrow from lone pair on <u>O</u> of OH⁻ / -OH to central C of 2-bromo 2-methylpropane if S_N2 mechanism shown	1
3(c)(ii)	nucleophilic substitution	1

36. M/J 21/P22/Q3(d)

3(d)	M1 more time (because the rate is lower)	1
	M2 C-Cl (bond) is stronger (than C-Br)	1

37. M/J 21/P22/Q5(b)(iii)

5(b)(iii)	<i>reagent</i> M1 NaCN or KCN	1
	<i>conditions</i> M2 ethanolic AND heat (under reflux)	1

38. M/J 20/P21/Q5(c)(ii)

5(c)(ii)	tertiary halogenoalkane	1
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39. M/J 20/P22/Q4(a)(i)-(ii)

4(a)(i)	Bromine / Br	1
4(a)(ii)	Ag ⁺ (aq) + X ⁻ (aq) → AgX(s) OR Ag ⁺ (aq) + Br ⁻ (aq) → AgBr(s)	1

40. M/J 20/P22/Q4(b)

4(b)(i)	CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ Br	1
4(b)(ii)	M1 primary/ 1° (carbo)cation formed is not very stable M2 EITHER (as) only one alkyl group exerting an inductive effect OR only one alkyl group so the charge is (more) localised on the C+	2

41. M/J 20/P22/Q4(c)(i)-(ii)

4(c)(i)	elimination	1
4(c)(ii)	C ₄ H ₉ Cl + NaOH → C ₄ H ₈ + NaCl + H ₂ O	1

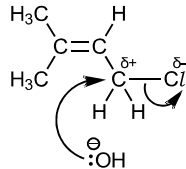
42. MAR 20/P22/Q2(e)(iv)

2(e)(iv)	substitution	1
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43. O/N 19/P22/Q3(c)(iv)

3(c)(iv)	M1: C ₂ H ₅ I reacts fastest AND C ₂ H ₅ Cl reacts slowest OR C ₂ H ₅ Cl < C ₂ H ₅ Br < C ₂ H ₅ I M2: C—I bond is the weak(est) AND C—Cl bond strong(est)	2
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44. O/N 19/P22/Q4(a)

4(a)	 M1: curly arrow from lone pair on O of OH⁻ to C of C—Cl M2: correct dipole on C—Cl AND curly arrow from C—Cl bond to Cl^{δ-}	2
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45. O/N 19/P22/Q4(e)(ii)

4(e)(ii)	potassium cyanide / KCN / sodium cyanide / NaCN	1
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46. M/J 19/P23/Q5(a)(ii)-(iv)

5(a)(ii)	a lone pair / electron pair donor	1
5(a)(iii)	(:)CN ⁻ / ⁻ (:)CN / cyanide ion	1
5(a)(iv)	Br (atom) is replaced (with / by CN / nitrile)	1

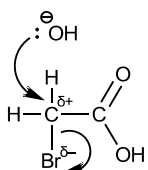
47. M/J 19/P23/Q5(c)

5(c)(i)	M1 Increasing reactivity from Cl → Br → I M2 Due to decreasing strength of C-X bond (from C-Cl to C-Br to C-I) OR Less energy needed to break C-X (from C-Cl to C-Br to C-I)	2
5(c)(ii)	M1 tertiary / 3° halogenoalkane M2 (carbo)cation / intermediate is stable M3 due to (3) electron releasing/donating methyl groups / + I groups (attached to central C) OR (positive) inductive effect of the (three) methyl groups /	3
5(c)(iii)	Any formula / name for any primary halogenoalkane i.e. 1-chlorobutane / 1-bromobutane / 1-iodobutane	1

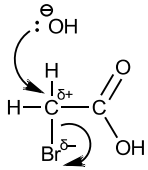
48. MAR 19/P22/Q4(c)(iii)

4(c)(iii)	nucleophilic substitution / S _N 2	1
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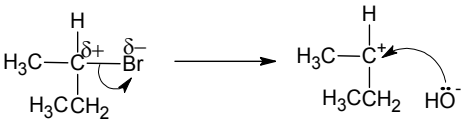
49. O/N 18/P21/Q4(b)(v)

4(b)(v)	 M1 lone pair on ⁻OH AND curly arrow from lone pair to C of C—Br M2 correct dipole on C—Br AND curly arrow from bond to Br	2
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50. O/N 18/P23/Q4(b)(v)

4(b)(v)	 M1 lone pair on OH^- AND curly arrow from lone pair to C of $\text{C}-\text{Br}$ M2 correct dipole on $\text{C}-\text{Br}$ AND curly arrow from bond to Br	2
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51. M/J 18/P23/Q4(c)

4(c)(i)	S = substitution N = nucleophilic	1
4(c)(ii)	 M1 = dipole and curly arrow M2 = intermediate carbocation IGNORE CH_2CH_3 shown as C_2H_5 M3 = OH^- with lone pair and curly arrow from lone pair to the 'C'	3

52. M/J 18/P23/Q4(d)

4(d)	$\text{C}_4\text{H}_9\text{Cl}$ slower (than $\text{C}_4\text{H}_9\text{Br}$) $\text{C}_4\text{H}_9\text{I}$ faster (than $\text{C}_4\text{H}_9\text{Br}$) $\text{C}-\text{Cl}$ bond stronger than $\text{C}-\text{Br}$ OR $\text{C}-\text{Cl}$ 340 $\text{C}-\text{Br}$ 280 $\text{C}-\text{I}$ bond weaker than $\text{C}-\text{Br}$ OR $\text{C}-\text{Br}$ 280 $\text{C}-\text{I}$ 240 1 mark for each point, max 3	3
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53. M/J 18/P23/Q4(e)

4(e)	ethanolic (instead of aqueous)	1
	hotter/higher temperature (heat under) reflux	1

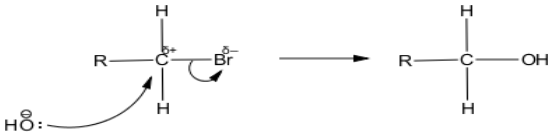
54. MAR 18/P22/Q4(b)

4(b)	elimination	1
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55. O/N 17/P21/Q3(a)

3(a)	<table border="1"> <thead> <tr> <th>reaction</th><th>reagent(s) and conditions</th><th>reaction type(s)</th></tr> </thead> <tbody> <tr> <td>1</td><td>aqueous / aq / dilute NaOH / KOH OR water</td><td>substitution OR hydrolysis</td></tr> <tr> <td>2</td><td>alcoholic / ethanolic NaOH / KOH</td><td>elimination</td></tr> <tr> <td>3</td><td>NaCN / KCN in ethanol / alcohol</td><td>substitution</td></tr> <tr> <td>4</td><td>aqueous / dilute H_2SO_4 / $\text{H}^+(\text{aq})$</td><td>hydrolysis OR substitution OR addition-elimination</td></tr> <tr> <td>5</td><td>acidified / H^+ (with) $\text{K}_2\text{Cr}_2\text{O}_7$ / $\text{Cr}_2\text{O}_7^{2-}$ (and distil) NOT reflux</td><td>oxidation OR elimination</td></tr> <tr> <td>6</td><td>acidified / H^+ $\text{K}_2\text{Cr}_2\text{O}_7$ / $\text{Cr}_2\text{O}_7^{2-}$ Fehling's / Tollens' / Benedict's (reagent)</td><td>oxidation</td></tr> </tbody> </table>	reaction	reagent(s) and conditions	reaction type(s)	1	aqueous / aq / dilute NaOH / KOH OR water	substitution OR hydrolysis	2	alcoholic / ethanolic NaOH / KOH	elimination	3	NaCN / KCN in ethanol / alcohol	substitution	4	aqueous / dilute H_2SO_4 / $\text{H}^+(\text{aq})$	hydrolysis OR substitution OR addition-elimination	5	acidified / H^+ (with) $\text{K}_2\text{Cr}_2\text{O}_7$ / $\text{Cr}_2\text{O}_7^{2-}$ (and distil) NOT reflux	oxidation OR elimination	6	acidified / H^+ $\text{K}_2\text{Cr}_2\text{O}_7$ / $\text{Cr}_2\text{O}_7^{2-}$ Fehling's / Tollens' / Benedict's (reagent)	oxidation	6
reaction	reagent(s) and conditions	reaction type(s)																					
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6	acidified / H^+ $\text{K}_2\text{Cr}_2\text{O}_7$ / $\text{Cr}_2\text{O}_7^{2-}$ Fehling's / Tollens' / Benedict's (reagent)	oxidation																					

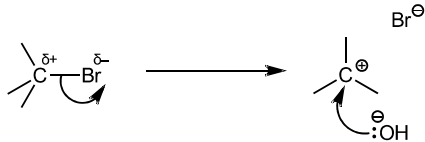
56. O/N 17/P21/Q3(b)

3(b)	 M1 lone pair on O of OH^- AND curly arrow from lone pair to C(—Br) M2 correct dipole on $\text{C}^{\delta+}\text{—Br}^{\delta-}$ AND curly arrow from bond to Br	2
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57. O/N 17/P21/Q3(c)(ii)

3(c)(ii)	S_{N} / nucleophilic substitution	1
	$((\text{CH}_3)_3\text{CBr}$ / tertiary halogenoalkane) forms a stable (carbo)cation / stable intermediate (as charge density on cation is reduced) OR (in) 1-bromobutane / primary halogenoalkane there is no (stable) (carbo)cation / intermediate formed	1
	(because) there are (3 /more) alkyl / methyl groups AND (+) I / (greater) inductive effect OR (because) there is only one / fewer alkyl / methyl group(s) (compared to reaction with 2-bromo-2-methyl propane / tertiary halogenoalkane) AND limited (+) I / (less) inductive effect	1

58. O/N 17/P22/Q4(b)

4(b)	 M1 correct dipole on $\text{C}^{\delta+}\text{—Br}^{\delta-}$ AND curly arrow from C—Br bond to Br M2 correct intermediate with + charge M3 curly arrow from lone pair on OH^- to C^+ of carbocation	3
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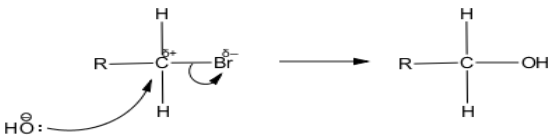
59. O/N 17/P22/Q4(c)(ii)

4(c)(ii)	S_{N} / nucleophilic substitution	1
	no (stable) (carbo)cation / intermediate is formed	1
	only one alkyl group / fewer alkyl / methyl groups (compared to reaction 2) AND limited (+)I / inductive effect / less electron donating (effect)	1

60. O/N 17/P23/Q3(a)

3(a)	<table border="1"> <thead> <tr> <th>reaction</th><th>reagent(s) and conditions</th><th>reaction type(s)</th></tr> </thead> <tbody> <tr> <td>1</td><td>aqueous / aq / dilute NaOH / KOH OR water</td><td>substitution OR hydrolysis</td></tr> <tr> <td>2</td><td>alcoholic / ethanolic NaOH / KOH</td><td>elimination</td></tr> <tr> <td>3</td><td>NaCN / KCN in ethanol / alcohol</td><td>substitution</td></tr> <tr> <td>4</td><td>aqueous /dilute H_2SO_4 / H^+(aq)</td><td>hydrolysis OR substitution OR addition-elimination</td></tr> <tr> <td>5</td><td>acidified / H^+ (with) $\text{K}_2\text{Cr}_2\text{O}_7$ / $\text{Cr}_2\text{O}_7^{2-}$ (and distil) NOT reflux</td><td>oxidation OR elimination</td></tr> <tr> <td>6</td><td>acidified / H^+ $\text{K}_2\text{Cr}_2\text{O}_7$ / $\text{Cr}_2\text{O}_7^{2-}$ Fehling's / Tollens' / Benedict's (reagent)</td><td>oxidation</td></tr> </tbody> </table>	reaction	reagent(s) and conditions	reaction type(s)	1	aqueous / aq / dilute NaOH / KOH OR water	substitution OR hydrolysis	2	alcoholic / ethanolic NaOH / KOH	elimination	3	NaCN / KCN in ethanol / alcohol	substitution	4	aqueous /dilute H_2SO_4 / H^+ (aq)	hydrolysis OR substitution OR addition-elimination	5	acidified / H^+ (with) $\text{K}_2\text{Cr}_2\text{O}_7$ / $\text{Cr}_2\text{O}_7^{2-}$ (and distil) NOT reflux	oxidation OR elimination	6	acidified / H^+ $\text{K}_2\text{Cr}_2\text{O}_7$ / $\text{Cr}_2\text{O}_7^{2-}$ Fehling's / Tollens' / Benedict's (reagent)	oxidation	6
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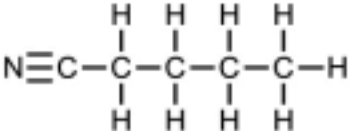
61. O/N 17/P23/Q3(b)

3(b)	 M1 lone pair on O of OH^- AND curly arrow from lone pair to C(—Br) M2 correct dipole on $\text{C}^{\delta+}\text{—Br}^{\delta-}$ AND curly arrow from bond to Br	2
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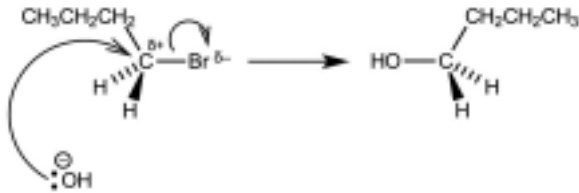
62. O/N 17/P23/Q3(c)(ii)

3(c)(ii)	S_{N} / nucleophilic substitution	1
	$((\text{CH}_3)_3\text{CBr}$ / tertiary halogenoalkane) forms a stable (carbo)cation / stable intermediate (as charge density on cation is reduced) OR (in) 1-bromobutane / primary halogenoalkane there is no (stable) (carbo)cation / intermediate formed	1
	(because) there are (3 /more) alkyl / methyl groups AND (+) I / (greater) inductive effect OR (because) there is only one / fewer alkyl / methyl group(s) (compared to reaction with 2-bromo-2-methyl propane / tertiary halogenoalkane) AND limited (+) I / (less) inductive effect	1

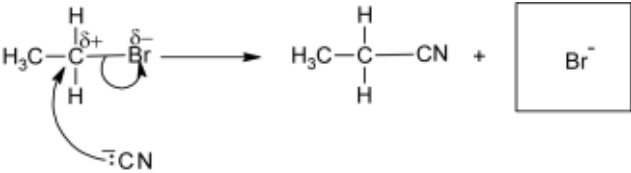
63. MAR 17/P22/Q3(a)(i)

3(a)(i)		1
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64. MAR 17/P22/Q3(b)

3(b)	 lone pair on O AND curly arrow from O to C of C—Br dipole on C—Br AND curly arrow from C—Br to Br product (butan-1-ol)	3
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65. M/J 16/P21/Q5(a)

5 (a)	 M1 = lone pair on C of CN^- AND curly arrow from lone pair to C of C—Br M2 = correct dipole on C—Br, curly arrow from C—Br bond to Br AND Br^-	[1]	[1]	[2]
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66. MAR 16/P22/Q4(e)

(e)	NaOH(aq)	[1]	[1]
			[11]

67. O/N 15/P21/Q4(a)(i)

4 (a) (i)	$\text{CH}_3\text{CH}_2\text{OH} + \text{HCl} \rightarrow \text{CH}_3\text{CH}_2\text{Cl} + \text{H}_2\text{O}$ or $\text{CH}_3\text{CH}_2\text{OH} + \text{PCl}_5 \rightarrow \text{CH}_3\text{CH}_2\text{Cl} + \text{HCl} + \text{POCl}_3$ or $\text{CH}_3\text{CH}_2\text{OH} + \text{SOCl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{Cl} + \text{HCl} + \text{SO}_2$	[1+1]	[2]
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68. O/N 15/P21/Q4(b)

(b) (i)	$\text{CH}_2=\text{CH}_2$ / ethane / C_2H_4 / CH_2CH_2	[1]	[1]
(ii)	White ppt / solid / suspension	[1]	[1]
(iii)	$\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$	[1]	[1]

69. O/N 15/P22/Q4(a)(i)

4 (a) (i)	Nucleophilic Substitution	[1]	[1]
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70. O/N 15/P22/Q4(a)(iv)

(iv)	Elimination	[1]	[1]
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71. O/N 15/P22/Q4(b)(ii)-(iii)

(ii)	propanoyl chloride / $\text{C}_2\text{H}_5\text{COCl}$	[1]	[1]
(iii)	$\text{CH}_3\text{CH}(\text{NHCO}_2\text{H}_5)\text{CH}_3$	[1]	[1]

72. O/N 15/P23/Q4(a)(i)

4 (a) (i)	$\text{CH}_3\text{CH}_2\text{OH} + \text{HCl} \rightarrow \text{CH}_3\text{CH}_2\text{Cl} + \text{H}_2\text{O}$ or $\text{CH}_3\text{CH}_2\text{OH} + \text{PCl}_5 \rightarrow \text{CH}_3\text{CH}_2\text{Cl} + \text{HCl} + \text{POCl}_3$ or $\text{CH}_3\text{CH}_2\text{OH} + \text{SOCl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{Cl} + \text{HCl} + \text{SO}_2$	[1+1]	[2]
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73. O/N 15/P23/Q4(b)

(b) (i)	$\text{CH}_2=\text{CH}_2$ / ethane / C_2H_4 / CH_2CH_2	[1]	[1]
(ii)	White ppt / solid / suspension	[1]	[1]
(iii)	$\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$	[1]	[1]

74. M/J 15/P22/Q3(b)(ii)

(ii)	KOH / NaOH	[1]	[2]
	ethanolic AND heat / reflux	[1]	